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PDA
Parenteral Drug Association

Technical Report No. 91

Post-Approval Change Management and Implementation for Biologics and Pharmaceutical Drug Products — A User's Guide

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Technical Report No. 91

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1.0 Introduction

The complexity of global post-approval change/variation (PAC) processes and the time required to receive their approval presents a challenge for industry when it is working to implement needed changes and operating with a focus on continual improvement. Industry benchmarking shows that receiving approval for a PAC requiring global approval can take between two and five years, and the time it takes from original submission can vary from six months to several years depending on the jurisdiction and review pathway ***(1)***. For new products, where the amount of commercial process or product data at the time of the initial submission is limited, this timing can significantly impact the ability to implement necessary process improvements identified as more data becomes available. In specific cases, when a change is needed to ensure the product remains safe, efficacious, and of high quality or provides additional capacity based on high product demand, the global approval timing can result in drug product shortages.

The current regulatory pathway to implement change is highly complex. For each jurisdiction in which a product is licensed, the respective regulatory agency typically requires a notification, based on its risk reporting category, or it must preapprove certain PACs for certain products. Differences in country- and region-specific requirements for PAC filings, filing procedures, and associated timelines can often make simultaneous implementation of changes problematic, necessitating a product be tested or produced using different processes based on where it will be distributed. Companies must consider this factor when deciding to implement a change.

These challenges were highlighted during the COVID-19 pandemic, as companies continued to manufacture lifesaving products despite the extreme strain on supply chains for raw materials, components, and manufacturing supplies. As alternative sources of supply or alternative materials were identified, the complexity of the global regulatory PAC process presented a significant challenge to product supply, highlighting the need for a harmonized approach to PAC implementation.

1.1 Purpose

This technical report expands on the latest concepts and tools for improving PAC management, as published in ICH Q12. It provides a practical guide for the pharmaceutical industry on how to implement PACs within the framework of an effective pharmaceutical quality system (PQS) and in alignment with ICH quality guidelines. It recognizes the challenges that the limited amount of global harmonization among regulators presents with respect to PACs and attempts to provide some insight on how industry can work with regulators to improve in this area. Though TR-91 aligns with ICH Q12, it also provides guidance for non-ICH regions. It suggests how to implement the tools described in ICH Q12 and provides examples of the concepts that can facilitate aligned industry understanding for PACs.

TR-91 highlights the benefits of some of the tools described in ICH Q12 that are already available in certain jurisdictions, specifically, PAC management protocols (PACMPs, or comparability protocols) and established conditions (ECs). These concepts were acceptable for use by several regulatory agencies prior to ICH Q12, that is, ECs and comparability protocols by the U.S. Food and Drug Administration (FDA) ***(11,12)*** and comparability protocols by the European Medicines Agency (EMA) ***(13)*** and WHO ***(4,5)***.

Section 10.0 (Appendix 1) and **Section 11.0** (Appendix 2) provide examples of PACs common in industry; these are presented as representative frameworks for consideration, presented as hypothetical PACMPs (or comparability protocols) that can be leveraged for similar changes. These examples, although not all-inclusive, do offer insights into the types of documentation and the scientific justification that may be required for implementation. They are intended to be illustrative, but they do portray how negotiation with regulators, where existing requirements may be disproportionate to risk given either knowledge gained through development or process experience, could be demonstrated using a PACMP approach.

1.2 Scope

The information in this technical report focuses on chemistry, manufacturing, and controls (CMC) changes that can be applied to PACs for both small-molecule and large-molecule pharmaceutical products.

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2.0 Glossary of Terms

2.1 Glossary

Approved Matters
A piece of information described in a dossier for an application which was approved pursuant to Japanese Ministry of Health, Labour, and Welfare, Article 14, Paragraph 1 or Article 19-2, Paragraph 1 of the Law ***(14)***.

Critical Process Parameter
A process parameter whose variability has an impact on a critical quality attribute and therefore should be monitored or controlled to ensure the process produces the desired quality ***(7)***.

Change Management
A systematic approach to proposing, evaluating, approving, implementing, and reviewing changes ***(9)***.

Continual Improvement
Recurring activity to increase the ability to fulfill requirements ***(15)***.

Control Strategy
A planned set of controls derived from current product and process understanding that assures process performance and product quality. The controls can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control ***(9).***

Critical Quality Attribute
A physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality ***(7).***

Established Conditions
FDA The description of the product, manufacturing process, facilities and equipment, and elements of the associated control strategy, as defined in an application, that assure process performance and quality of an approved product ***(11).***

ICH ECs are legally binding information considered necessary to assure product quality. As a consequence, any change to ECs necessitates a submission to the regulatory authority ***(6).***

Innovation
The introduction of new technologies or methodologies ***(9).***

Knowledge Management
Systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components ***(9)***.

Notification
The submission to a regulatory authority of a change in established conditions, or a change that has the potential to have a minor or moderate impact to product quality, safety, and/or efficacy, that does not require a regulatory authority approval prior to implementation.

Pharmaceutical Quality System
Management system to direct and control a pharmaceutical firm with regard to quality ***(9, 15)***.

Prior Approval
Change to an approved established condition, or a change that has the potential to have a major impact to product quality, safety, and/or efficacy, that requires regulatory review and approval prior to implementation.

Product Lifecycle
All phases in the life of a product from the initial development through marketing until the product's discontinuation ***(7)***.

Product Lifecycle Management
The Product Lifecycle Management (PLCM) document outlines the specific plan for product lifecycle management that includes the ECs, reporting categories for changes to ECs, PACMPs (if used), and any post-approval CMC commitments. Its purpose is to encourage prospective lifecycle management planning by the marketing authorization holder and to facilitate regulatory assessment and inspection. The PLCM document should be updated throughout the product lifecycle as needed ***(6)***.

Product Realization
Achievement of a product with the quality attributes appropriate to meet the needs of patients, health care professionals, and regulatory

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authorities (including compliance with marketing authorization) as well as internal customers' requirements ***(9)***.

Post-Approval Change Management Protocol (or Comparability Protocol)

FDA A comprehensive, prospectively written plan for assessing the effect of proposed CMC postapproval change(s) on the identity, strength, quality, purity, and potency of a drug product or a biological product (i.e., product), as these factors may relate to the safety or effectiveness of the product (i.e., product quality) ***(11)***.

ICH A regulatory tool that provides predictability and transparency in terms of the requirements and studies needed to implement a change as the approved protocol provides an agreement between the marketing authorization holder and the regulatory authority ***(6)***.

Process Capability

Ability of a process to realize a product that will fulfill the requirements of that product. The concept of process capability can also be defined in statistical terms ***(12)***.

Quality

The degree to which a set of inherent properties of a product, system, or process fulfills requirements ***(8).***

State of Control

A condition in which the set of controls consistently provides assurance of continued process performance and product quality ***(9)***.

2.2 Abbreviations

API
Active pharmaceutical ingredients

CBE
Changes Being Effected

CMC
Chemistry, manufacturing, and controls

CPP
Critical process parameter

CQA
Critical quality attribute

EC
Established conditions

EMA
European Medicines Agency

FDA
U.S. Food and Drug Administration

cGMP
Current good manufacturing practices

ICH
International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use

PAC
Post-approval change

PACMP
Post-approval change management protocol

PAS
Prior Approval Supplement

Ph. Eur.
European Pharmacopoeia

PQS
Pharmaceutical quality system

USP
United States Pharmacopeia

WHO
World Health Organization

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3.0 Background and Landscape for PACs

The importance of PAC and the need to overcome these myriad challenges has led many international organizations, including the Parenteral Drug Association (PDA), to develop recommendations and proposals to address the situation:

PDA established the Post Approval Change; Innovations of the Availability of Medicines (PAC iAM) program that has developed numerous papers, including this technical report.

The European Federation of Pharmaceutical Industries and Associations and International Federation of Pharmaceutical Manufacturers and Associations have issued position papers on PAC, calling for regulatory convergence and proposing a relevant framework ***(1, 2)***.

One Voice of Quality for PAC, an initiative sponsored by the chief quality officers of more than 30 global pharmaceutical companies, has been developing risk-based recommendations to help reduce the global complexity associated with PACs ***(3)***.

The World Health Organization (WHO) has provided guidance, suggesting pathways and requirements for PACs ***(4, 5)***.

The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) has continued its push for harmonization by publishing *Quality Guideline Q12: Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management,* which focuses specifically on PACs ***(6),*** while building upon established ICH Quality Guidelines Q8-Q11 ***(7–10)***.

The ICH Q12 guidance, which describes key concepts necessary to streamline the PAC process, is recognized as a significant step forward. In addition to the implementation of ICH Q12, there are areas where work is still needed to achieve full harmonization, including:

- Agreement on the change categories among regulatory authorities
- Harmonization on filing procedures and timelines within ICH regions
- Inclusion of non-ICH regulatory authorities to converge on the application of these principles

By fully implementing ICH Quality Guidelines Q8–Q11, manufacturing firms can develop an effective internal PQS sufficient to optimize the management of PACs. Demonstrating the effectiveness of the PQS would allow a reduction in the regulatory reporting burden for changes with a lower level of risk, as they would be managed through a firm's PQS, by:

- Ensuring ICH Q8–Q12 are implemented and leveraged as building blocks for PAC management
- Working with regulators to drive consistent reporting categories
- Developing a robust PQS system
- Enhancing transparency and trust between companies and regulators to identify when a firm's PQS system is sufficient to allow certain types of lower-risk change to be implemented without reporting

In keeping with ICH Q10 and Q12 guidances and with a robust PQS system in place, PACs requiring prior approval by regulatory authorities would be limited to changes assessed as having the potential to have a major impact on product quality and/or patient safety. Minor-to-moderate changes would be made with notification to the regulatory authorities, and low-risk changes would be managed under a firm's PQS. The benefits of this paradigm shift are depicted in **Figure 3.0-1**.

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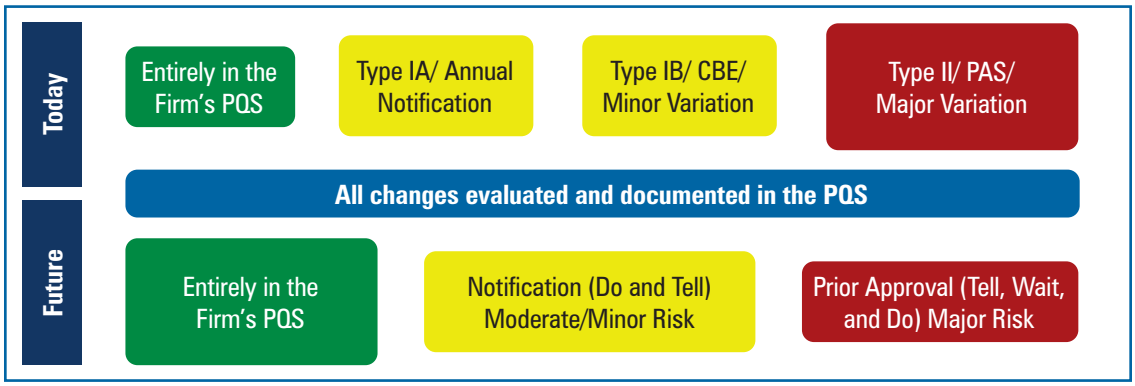

Figure 3.0-1 Scientific and Risk-Based Approach to Categorization of Changes

4.0 Key Elements of a Pharmaceutical Quality System for PAC Management

An effective PQS is the foundation upon which PAC management and implementation are established. PQS effectiveness in relation to risk-based change management is a core concept within the ICH Q12 framework. Several PQS elements are needed to establish effective PAC management: management review, a corrective action and preventive action (CAPA) program, a change management system, and process performance and product quality monitoring ***(3)***. Importantly, knowledge management and quality risk management are key enablers of a firm's PQS.

Having an established PQS that is executed effectively and consistently within a firm will provide confidence to the regulatory agency that changes are supported by data obtained through the application of patient-centric, scientific, risk-based principles. A well designed and effective PQS is an essential foundation for trust between regulatory authorities and industry that will facilitate harmonization on the categorization of changes. It also should allow more PACs to be managed under the firm's PQS without requiring prior review and approval by regulatory agencies ***(3)***.

When using contract manufacturing organizations (CMO) or contract development and manufacturing organizations (CDMO), a firm needs to ensure that appropriate contractual agreements are in place for PACs occurring at the CMO or CDMO and that they maintains an effective PQS ***(16)***. The same principles for lifecycle management and reporting responsibility apply.

4.1 Management Review

A firm's senior management is responsible for ensuring that appropriate systems and processes are in place to manage changes effectively. In accordance with ICH *Quality Guideline Q10: Pharmaceutical Quality System,* these responsibilities include:

- Defining and communicating roles within the change management system
- Ensuring timely and appropriate communication to address potential quality issues
- Providing adequate resources to ensure that changes are made based on sound scientific and risk-based judgments
- Maintaining awareness of the long-term product lifecycle management strategy and ensuring that appropriate controls are in place
- Developing and fostering the desired quality culture at all levels throughout the firm to safeguard the patient's access to high-quality medicines

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Quality culture is a set of behaviors, beliefs, and values of an enterprise that serve as a guide to decision-making. In the ideal state, a quality mindset and quality-focused behaviors are embedded into all functions of daily work and result in positive patient outcomes. In an organization with a well-established quality-culture environment, employees not only follow quality guidelines but consistently see others taking quality-focused actions, hear others talk about quality, and understand the importance of quality in everyday activities ***(17)***.

For sustained effectiveness, the PQS needs to be embedded alongside a firm's culture of quality for it to properly perform its function and be considered reliable. Management plays an important role in nurturing key elements of a mature quality culture, including a focus on continuous improvement, demonstration of leadership, communication of a common vision, and investment in employee development and education. Training is essential to ensure a proficient and knowledgeable quality-oriented workforce.

4.2 Corrective Action and Preventive Action Program

A properly functioning CAPA system identifies not only when actions are needed as a result of an event that has occurred, but also operates proactively to improve based on weaknesses identified prior to an event occurring. Ideally, the majority of all actions would be preventative. The CAPA system is often where the initial need for change is identified and where the PAC process begins. Having a strong CAPA program is a critical component of any PQS system, which will lead to better process understanding, process control, and determining when action is needed.

An effective CAPA program monitors and manages unintended risks and consequences of PACs for internal and outsourced operations. It should enable identification of root cause(s) so that appropriate actions can be taken to correct problems and prevent their recurrence. The program should also monitor and verify the effectiveness of any CAPAs associated with PAC initiatives.

4.3 Change Management System

While most changes made have no impact on the regulatory dossier, the importance of having a system that appropriately evaluates changes and their impact on the manufacturing processes and product cannot be emphasized enough. The change process must be able to adequately evaluate the change, the level of risk to the process and product, and the need for a PAC to be filed. Tools, such as those described in ICH Q10 and Q12 and in the PIC/S *Recommendation*: *How to Evaluate/Demonstrate the Effectiveness of a Pharmaceutical Quality System in relation to Risk-based Change Management,* focus on the relevance of risked-based change management**.**

The PIC/S recommendation provides a checklist to evaluate the effectiveness of a firm's PQS in relation to risk-based change management and acknowledges that the level of effort and associated documentation should be commensurate with the level of risk the change entails. This, in turn, should be used to justify the categorization and filing requirements for that change ***(18).*** To ensure that the underlying risk level is well understood, the change management process needs to be robust enough to capture the impact of any proposed change. Therefore, it must support a knowledge-based assessment and review of the change, ensuring that the final scientific justification supports the ultimate filing decision to the applicable regulatory agency.

A structured, well-organized change management process is essential for a successful PAC process. It must cover critical quality attributes (CQA), critical process parameters (CPP), product lifecycle management (PLCM), and control strategy and include creating ECs as appropriate, among other elements (see **Section 4.1**). Such an approach ensures that relevant quality aspects are considered, and possible quality impacts are thoroughly assessed, documented, and communicated to relevant stakeholders, including regulatory bodies.

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4.4 Process Performance and Product Quality Monitoring

A comprehensive program to monitor the manufacturing process performance and product quality should be in place at every manufacturing facility to ensure that the process continues to perform in a state of control and in keeping with the validated state. A sound monitoring plan allows a firm to establish the baseline process performance and detect early warnings of atypical process performance. Having some process monitoring data before and after the change is an essential component of the overall qualification of any change. In essence, product and process monitoring results serve as the scientific basis for improvement of the manufacturing process and the control system. Typically, after a process change is put in place, monitoring frequency is increased to confirm that the process continues to perform as expected. For example, documentation of any deviations from trend limits and/or adverse trends with respect to process, in-process, release, and/or stability testing parameters should be carefully assessed in line with the quality system. These deviations should be investigated for root cause and evaluated to ensure the change has no adverse impact on product quality.

Effective change management is fundamental to ensuring that PACs are controlled in an environment characterized by data-driven, scientific, and risk-based assessments. In an effective change management environment, all aspects of the product and process are managed to ensure that there are no unintended consequences and that product quality and process performance are maintained in a state of control.

4.5 Quality Risk Management and Knowledge Management

Quality risk management (QRM) and knowledge management (KM) are key enablers of an effective PQS. An organization that strives to have a strong, quality-oriented culture will use QRM and KM to guide the identification and proactive management of PACs. *PDA Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations* provides a comprehensive description of the QRM elements and tools aligned with ICH Q9. In addition, annexes to TR-54 supply examples of QRM application for manufacturing processes and equipment qualification that may be relevant for PAC management ***(19).***

An effective QRM system provides a sound framework that ensures systematic and proactive risk-based and data-driven decision-making is being employed for all PACs. QRM should enable innovation by allowing faster implementation of changes that reduce the risk of quality failures and manufacturing problems and improve process capability. An effective QRM system can also separate changes that require regulatory reporting or notification from changes that can be managed completely within the PQS. Effective QRM enables a firm to demonstrate an understanding of the existing production process and of the risks, associated impacts, and mitigation of those risks in that process.

An effective KM system leverages historical and recently identified product and process knowledge, including innovation and technology advancements. It also considers knowledge gained from process monitoring, deviations, trends, complaints, recalls, annual product or product quality review, audits, inspections, and management reviews. Effective KM ensures that the right knowledge is used by the right people at the right time, allowing for effective risk-based decision-making and formal product and process risk assessments. A systematic, transparent, and well-understood KM system ensures that changes are based on scientific rationale and can facilitate implementation of identified changes accordingly.

Several FDA, Health Canada, and WHO guidance documents on CMC changes in a post-approval situation demonstrate how product and process knowledge and effective QRM can be leveraged to establish lesser reporting categories for changes ***(4, 5, 11, 20).*** FDA consistently states in guidances its intent to apply the principles of quality risk assessment and risk management in accordance with ICH Q9 during their review. The Agency recognizes that, in order to keep pace with the many advances in quality management practices in manufacturing and enable it to more effectively allocate its limited

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regulatory resources, implementing a risk-based approach for regulating pharmaceutical manufacturing is necessary ***(21).*** Accordingly, one FDA guidance allows that changes demonstrated to have a minimal impact to a product can be filed as an annual reportable item ***(22).*** Likewise, the Health Canada and WHO guidances, which provide conditions to be fulfilled for specific change types to help guide firms on how to assess risk and categorize changes accordingly, include the option of annual notification for minimal impact. In light of this, the principles of KM and QRM can be equally applied in both ICH and non-ICH regions and should be recognized by global regulatory authorities.

5.0 PAC Process Tools to Enhance Regulatory Flexibility

Maintaining GMP, combined with continuous learning, improvement, and in-depth knowledge about the production process and its CPPs and/or ECs, contributes to ongoing optimization. This optimization often results in changes to the product files submitted to and approved by the authorities, aiming for a new desired state. Regulatory tools should be used to enhance the management of these PACs in a way that encourages innovation and continuous improvement. Doing so may help a firm avoid time-consuming, iterative, and expensive processes and, instead, encourage smooth, straightforward interaction and transparent communication with regulatory authorities.

When deciding to implement a PAC, thereby aiming for a new desired state, the tools suggested in **Table 5.0-2** may prove helpful when writing the documentation and scientific justification. The use of tools and points for consideration, such as those identified in **Table 5.0-2**, can aid in identifying and presenting, at the outset, the most convincing and thorough documentation required for the regulatory approval process. However, while the use of a PACMP as discussed in ICH Q12 (see **Section 4.0**) and presented in **Table 5.0-2** continues to gain acceptance, its use is not currently accepted in all countries.

The decision about change implementation should focus on potential product impact and maintenance or improvement of quality and compliance. However, the development of a business case, as recommended under "Defining Scope of Change" in **Table 5.0-2**, can be a helpful tool that serves solely as internal documentation and initial justification of the change. The business case should clearly lay out pros and cons regarding cost versus benefit and impact on product quality. Other areas could potentially be impacted as well and should be considered, such as other manufacturing processes, submitted or drafted regulatory applications, quality or business procedures, functions or departments, collaborating partners, and stakeholders.

Subject matter experts are key stakeholders, and employee knowledge and insight are crucial to a successful PAC process. By obtaining increased process and product understanding, including science- and risk-based approaches for stability studies to support the evaluation of the changes, critical sources of variability can be identified and appropriately managed. By assessing potential impacts of the change, relevant mitigation strategies can be developed.

Written rationales justifying the changes should be based on the assessment of different parameters, for instance, impacts to CQAs, ECs (if applicable), and CPPs. When relevant, the process risk assessment should link process parameters and critical material attributes to CQAs, thereby defining the CPPs and linking them to an acceptable, scientifically justified range for the different process parameters. The type of change and the type of operational flexibility required should be considered; some changes may benefit from establishing a predefined design space. Design space is defined in ICH Q8(R2) as "[t]he multidimensional combination and interaction of input variables (e.g., material attributes) and process parameters that have been demonstrated to provide assurance of quality" ***(7).*** Working within

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an established design space offers the option of implementing changes and handling the PAC process within the PQS, thereby providing increased flexibility for the implementation of changes.

The control strategy should be designed and implemented based on quality risk management, which is a valuable tool to reflect and evaluate the different steps, by linking the process to the established CPPs and understanding impacts and/or changes to CQAs. The control strategy should be reassessed as part of the PAC process.

Table 5.0-9 Tools and Considerations for Decision-Making for PAC

	Defining the Scope of Change	Understanding	Risk Assessment	Regulatory Assessment	Definition	Approval Strategy	Implementation
Process	• What is the scope-background? • What is rationale-justification? • What is the benefit? • What other solutions have been evaluated?	• Have Established Conditions (ECs) been established/filed? • Have CQAs and CPPS been established? • Has a control strategy been established? • Has a PACMP been considered?	• Determined categorization by performing a risk assessment of the change related to patient safety, product quality or efficacy • If the risk is moderate to low -> lesser filing may be considered • If the risk is high -> a prior approval submission is likely	• Q12 PAC procedure or other procedure? • Are ECs, CPPs, and/or CQAs impacted? • If NO and has minimal impact -> change may be handled in the PQS, specific jurisdiction notification may be required, impact to CTD needs to the assessed • If YES-> assess for specific filling type per jurisdiction	• Define design space, as appropriate • Defined performance space, as appropriate • Define qualification/validation activities, as applicable • Create a stakeholder assessment -who might be affected and who has to be involved in the process?	• Define approval strategy for ICH countries • Define approval strategy for countries following ICH guidelines • Define approval strategy for all other countries • Contact authorities according to strategy • Prepare for implementation	• Create/update PACMP, as required • Submit protocol to authorities • Implement change • Verify implementation & validation • Submit report to authorities
Tools	• Business Case • Timeline • System impact assessment	• Product and process understanding • Process risk assessment (QRM) • Control strategy	• Change Risk Assessment • Quality Risk Assessment	• PQS • Change Categorization	• Quality by Design (QbD) • Validation Protocol • Stakeholder Assessment	• Approval strategy • Change categorization	• PACMP • Validation protocol and report

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5.1 Established Conditions

EC is an important term and concept in evaluating change and, while ICH Q12 works to bring more definition to the term, the concept has been in place for some time.

5.1.1 Background

ICH Q12 defines ECs as legally binding information considered necessary to assure product quality. Consequently, any change to an EC necessitates a submission to the regulatory authority**.** ICH Q12, which the FDA officially adopted in May 2021, further considers ECs to be the elements of the pharmaceutical development activities that result in an appropriate control strategy ***(23)***.

Previously, FDA defined ECs as the description of the product, manufacturing process, facilities and equipment, and elements of the associated control strategy, as defined in an application, that assure process performance and quality of a product ***(11)***. Japan has a similar concept, called "approved matters," that refers to an item of information described in a dossier for an application that was approved by the regulatory agency ***(14)***. Health Canada also leverages a similar concept, requesting that market authorization holders file and maintain a Certified Product Information Document. That document constitutes part of the approval package and provides an accurate record of key quality information for the product proposed for marketing at the time the approval is issued. Thereafter, it serves as an official reference document during post-approval inspections and PAC evaluations performed by Health Canada ***(20)***. Hence, the expectation of regulators to assess changes for the potential impact on product quality, safety, and efficacy has always existed.

Changes that could potentially have a major impact on product quality usually have the highest reporting category. As a term and a concept, EC may guide both the firm and the regulators in predefining the parameters that are more likely to impact the product. ICH Q10 describes the control strategy as "a planned set of controls, derived from current process and product understanding, that assures process performance and product quality," which may include CPPs and CQAs ***(9)***. Some elements of the control strategy may be managed in the applicant's PQS and not necessarily detailed in the dossier or application. Others may be detailed in the dossier but considered to be supportive information rather than ECs.

In the ICH Common Technical Document (CTD), ECs are already defined in *Module 3: Quality*. As per ICH Q12, all regulatory dossiers contain a combination of ECs and supportive information. Although identification of ECs with appropriate justifications might allow for increased flexibility in reporting PACs, explicitly identifying ECs for a given product is not necessarily mandatory. If the market authorization holder chooses to identify ECs, they should clearly identify the elements of Module 3 that they consider to be ECs and those they consider to be supportive information. The rationale for the ECs should remain in the appropriate CTD section. Maintaining a tabular summary of the ECs by section and providing those in the dossier, explicitly naming them as such, is recommended ***(24)***. Another tool for defining and tracking ECs is the use of a PLCM strategy document; the PLCM document should be maintained in the firm's change management system and in CTD Module 3.2.R, as per ICH Q12.

In addition to explicitly identifying ECs, the firm should propose the type of filing that will be provided should a future change impact an EC. Once a submission is approved and the ECs are agreed upon by the firm and the regulatory agency, they are considered legally binding, and any changes to the EC must be reported to the regulatory agency. The reporting mechanism or filing type may vary, however, and the supporting data for a change may be part of the subsequent filing or may be maintained in the firm's PQS (see **Section 5.2**).

5.1.2 Benefits of Defining Established Conditions

ECs can be defined as part of the initial product development or through a review of product knowledge obtained over the product's lifecycle. They should be limited to the elements found essential to controlling the quality of a drug product based on scientific and risk-based justification. Careful evalu-

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ation of the product and manufacturing process will assist in focusing on the parameters critical to quality control, which will reduce the number of non-value-added PACs that require prior approval by regulatory agencies. ECs will depend on the complexity of the product and the capabilities inherent to the manufacturing process and manufacturing site. ECs may vary by dosage form and drug substance but, fundamentally, they should not vary by region or country. Regulatory convergence across jurisdictions or regulatory reliance under mutual agreement is required to fully realize the benefit.

Clearly identified and justified ECs in an application ensure that the firm and the regulatory agency are aligned on the criticality of a process parameter or quality attribute. As a result, there should be agreement on the necessity for subsequent submissions and the filing category for changes to any given EC. While a change to an EC will require a filing with a regulatory agency, the type of filing can depend on the risk assessment and the process and product knowledge available. The risk may be product- and/or process-dependent and, importantly, how the risk is assessed is dependent on the effectiveness of the PQS in place. Ultimately, the same type of change may require different filing strategies for different products. This further highlights the importance of product and process knowledge and an effective PQS to fully realize the benefits of identifying ECs. For example, if the risk is considered low because the change will have little or no impact on product quality, the change may be made through a notification; if the risk is considered high due to a potentially high impact on product quality, prior approval of the change may be required. The documentation of these assessments and supporting data must live in the firm's PQS. When a prior approval supplement or variation is required, a PACMP may be leveraged to align the planned supportive studies and expected outcomes. This would lower the filing category and reduce the review scrutiny for the submission, which allows for implementation of the change more rapidly and efficiently (see **Section 6.0**).

ICH *Quality Guideline Q12 Annexes: Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management* illustrate how development approaches can impact product understanding and, ultimately, the identification of ECs ***(25)***.

Sections 10.0 (Appendix 1) and **11.0** (Appendix 2) provide examples of ECs and the PACMPs developed to revise them.

In line with ICH Q10, if a firm can leverage the utility of a robust PQS system—one that incorporates KM and sound quality risk and change management strategies—the firm can gain increased flexibility to change ECs as more process and product knowledge is developed. A robust PQS will also enhance the firm's understanding of the product and its manufacturing process which, over time, will allow the firm to critically analyze ECs and reevaluate their potential impact on product quality. That critical evaluation will afford the firm the potential to remove ECs and negotiate lower filings for changes associated with existing ECs. This will then facilitate process and facility improvements.

In the current environment, firms and regulatory agencies frequently experience a lengthy submission and approval process. By predefining ECs and gaining regulatory agency agreement, firms can reduce the number of PACs that require regulatory agency submission or approval. The filing categories will be reduced for those changes that do require submission. This would both enable faster implementation of changes to the drug product and mitigate drug shortages related to lifecycle management. As they support continual improvement, PACs can result in decreased product variability, reduced supply shortages, and increased manufacturing efficiency, as well as facilitate the introduction of innovation.

5.1.3 ECs versus Non-ECs and Change Management

All changes are managed through the firm's change management system, that is, within the PQS, as illustrated in ICH Q12. This guidance suggests that any parameter that needs to be controlled to ensure product quality (i.e., a CPP or process parameter where impact on quality cannot be reasonably excluded) should be considered an EC. ECs are intended to provide understanding and agreement

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between the firm and the regulatory agency on the parameters necessary to assure product quality; these parameters can change with knowledge gained throughout the lifecycle of the product. The focus should be on understanding and identifying ECs and appropriately linking them to an agreed-upon post-marketing plan and regulatory oversight. As such, process parameters, which should be assigned based on criticality to the process and its control strategy, can be considered ECs and can be defined with regulatory agencies based on process and product knowledge. Yet, not all data and/or parameters described in a filing will necessarily be ECs; they may be detailed in the dossier but only as supportive information. To define the changes that may require a regulatory agency filing, it is critical both to have explicitly defined ECs and to understand what elements are not. Ultimately, appropriately identifying ECs (and therefore non-ECs) can be a useful tool for harmonization in regions not participating in ICH when assessing the impact of PACs.

5.2 Post-Approval Change Management Protocols

The PACMP tool described in ICH Q12 enables planning and implementation of changes to ECs and/or CPPs, and thereby to regulatory filings, in an efficient and structured manner. The concept of the PACMP is not new. The FDA and EMA have accepted comparability protocols for several years, and WHO guidances discuss the use of comparability protocols as a tool to help expedite the distribution of product post-change ***(4, 5, 12, 13)***. The PACMP (or comparability protocol) approved by the relevant health authority provides an agreement between the firm and the regulatory agency and describes changes with a level of detail complying to the complexity of the change, including risk assessment, control strategy, and other pertinent elements. Where possible, a description in the original protocol of the types of changes that might be considered allowable, and/or how they will be communicated to the agency post-approval, should be included. Once a PACMP is approved, when the proposed study design or acceptance criteria changes, an update of the protocol should be submitted for implementation as agreed upon in the original protocol. If changes are needed due to unanticipated or unforeseen changes in plans, prior approval of a new or revised PACMP by the regulatory agency may be required.

A firm's processes may include internal guidance for filing initial market registrations, including templates for PACMPs. In general, PACMPs can include a description of the product's current state, the change to be made, the ECs and/or the product quality attributes that will be affected, the defined testing and acceptance criteria to establish comparability of the changed product, and the proposed category for filing the change.

Typically, a PACMP describes a single change or a group of related changes that may be made at the same time, for example, a scale-up of a purification process that involves larger equipment and associated changes to process parameters. In addition, PACMPs can be used to describe the same change or changes being implemented across multiple products, such as a new drug product container–closure system being implemented across a portfolio of products. In this case, the same PACMP could be filed for multiple applications, and subsequent submissions to provide the final data could be filed for each individual application, including a combination of product-specific and non-product-specific data.

Contents for PACMPs may vary from one firm to another and from one type of change to another, but as a best practice, all PACMPs should contain the following elements:

- Description of the current state and the proposed change as well as a justification or scientific rationale for the change (e.g., issues or disadvantages with the current state and benefits or advantages with the proposed state)
- Proposed submission category or submission type for the subsequent filing that includes the data, clearly indicating which sections of the registration will be impacted
- Detailed list of change(s)

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- Risk assessment (i.e., risk-based approach) to explain the potential impact of changes to product quality
- Study protocol describing how the potential impact(s) would be assessed to adequately demonstrate comparability of the product quality before and after the change
- List of predefined acceptance criteria that must be met before implementing the change
- Validation or qualification of the changed method/process/equipment to demonstrate that all acceptance criteria are met, as required, based on the risk level of change to the product quality
- Additional process characterization and validation studies at small-scale, as required, based on the specific change or, minimally, an assessment to confirm that the existing process characterization and validation studies adequately support the new process conditions
- List of post-change commitments, if any, including any stability studies to demonstrate accelerated and/or long-term stability of the product with the proposed change
- Anticipated timeline for implementation of the change
- Description of how the approved PACMP will be modified, if needed, due to additional knowledge gained during the actual execution

PACMPs may only be accepted by a limited number of regulatory agencies; however, harmonization of these best practices across ICH regions and some non-ICH regions would greatly improve the potential for using use these tools to facilitate the efficient implementation of PACs.

PACMPs may save firms considerable time and resources to effectively manage changes and provide regulators greater insight into planned changes, which would facilitate the review of post-change filings. This could alleviate constraints in the global supply chain, enabling a continuing supply of drug products to patients in a timely manner. Global acceptance by regulators, with a defined mechanism in local jurisdictions for submission of PACMPs, is needed to fully realize the benefits that PACMPs can offer.

Examples of PACMPs are provided in **Section 10.0** (Appendix 1) and **Section 11.0** (Appendix 2).

5.3 Product Lifecycle Management

The PLCM can be described in a strategy document that describes how the firm will:

- Monitor the process and product attributes to ensure that both remain within the validated state in accordance with the product control strategy (e.g., continued process verification)
- Manage and report changes to ECs
- Execute PAC filings that are planned as part of the product lifecycle

A PLCM strategy document may include the following sections:

- Summary of the product control strategy, e.g., a description of the CQAs for drug substance and drug product and the controls in place for raw materials, excipients, process steps, and final drug substance and drug product that ensure CQAs are maintained within acceptable ranges
- A table describing the CTD sections that contain ECs, the ECs in each of these sections, and the reporting categories that will be used if those ECs change
- PACMPs that may be used for future changes
- Post-approval commitments from the initial registration or subsequent supplements and the timelines for completion

A PLCM strategy document may provide an effective tool to help manage the commercial lifecycle of a product. It can be filed with the initial registration, or it may be filed as a post-approval variation.

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The document would also prove helpful in the firm's records as a repository of the commitments to the regulatory agency that must be maintained throughout the commercial lifecycle. The PLCM strategy document can also be used to communicate to the regulatory agency the elements of the registration that will be maintained during the commercial lifecycle, as well as the intended reporting categories and timing estimates. As post-approval variations are submitted, an updated PLCM should be included that describes the updated ECs, control strategy, reference to PACMPs, post-approval CMC commitments, and any other pertinent information about the change.

6.0 Global Lifecycle and PAC Management

6.1 ICH Regions

The impact of ICH in harmonizing guidelines and aligning ICH member regions has provided significant advantages in both the PLCM and the management of PACs. The continuing challenge is the wide diversity of expectations of non-ICH regions in both the content and timing of PAC submission approvals. Some common characteristics leveraged in ICH regions that enable efficient PACs include:

- Clear guidance for filing changes, including reporting categories based on risk
- Acceptance of science- and risk-based approaches for developing data to support PACs, such as bracketing and/or matrixing of process validation and stability batches
- Potential acceptance of limited stability data when justified, such as three or six months of real-time data and three months of accelerated data, to support major changes, and less or no stability data for minor changes
- Well-defined timelines for review and approval of PACs, in the range of six months for a major change and shorter periods for minor to moderate changes
- Relatively fewer requests for additional documentation not included in CTD Module 3, such as declarations or GMP certificates, although the latter may be needed for addition of a new facility

6.2 Non-ICH Regions

Regions not participating in ICH frequently observe fewer of the characteristics that enable PACs to be managed consistently across multiple regulatory authorities. Examples seen in some non-ICH regions include:

- Lack of available post-approval guidance
- No or very limited provision for a differentiation in filing level, such as allowing for providing notifications rather than submissions for lower-risk changes or recognizing risk-based scientifically valid strategies, such as bracketing or matrixing of process validation and stability data
- Requirements for extensive amounts of drug product stability data to support even minor changes or drug substance changes, for example, up to the full shelf life
- Requirements for extensive ancillary documentation to support PACs, such as declarations, certificates of various types, and legalizations

In 2020, PDA completed a survey of industry on this topic. According to survey results, half of the respondents indicated that the authorities that take the most time to approve prior-approval supplements require more than a year to grant such approval. Some markets were reported as having extremely long review times for changes, for example, five years. Some markets add extra requirements, such as implementing product changes immediately upon that country's approval, which is not always feasible given the manufacturing schedule or lag in approval by other jurisdictions. The survey also

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indicated some jurisdictions may not allow for the simultaneous review of multiple changes, instead requiring each change to be submitted for agency review individually, in sequence ***(26)***. These types of requirements make it extremely difficult to file global PACs, which can make it very challenging to initiate or even propose process improvements. In some cases, this can impact product supply.

When considering the supply chain, extended approval times for products with limited expiry dating can create significant challenges, as it may not be possible to build up sufficient inventory levels. For example, a process or facility change that requires modifications that, when made, no longer allows for the product to be manufactured using the original process. Once the change is made, the inventory of product using the original process is set and, if approval takes an extended time (in some cases years), the chance the inventory will be depleted or will expire before the change is fully approved increases significantly, impacting product availability and affecting the patient. The WHO guidances on biotherapeutics and vaccines provide good examples of changes and apply a risk-based approach to PACs, similar in format to the Health Canada *Post-Notice of Compliance (NOC) Changes: Quality Document* ***(20)***. The WHO publications are intended to serve as a guide for establishing national requirements for the regulation of PACs to biotherapeutic products and vaccines. In jurisdictions that do not offer their own guidance, these guidelines can be helpful in applying a risk-based approach to the PAC requirements and approval timing. In these cases, the guidance(s) should be referenced during the PAC process. Also, the WHO PAC guidelines strongly encourage national regulatory authorities to establish requirements commensurate with their own regulatory capacity, experience, and resources and to work toward minimizing any negative impact on access to approved products ***(4,5)***. National regulatory authorities are encouraged to consider establishing procedures for the expedited approval of changes based on previous expert review and approval of the same changes by the national authorities of the other countries where those products are licensed or, alternatively, based on the decision of a recognized regional regulatory agency.

Where there is a potential for the timing of change approval in non-ICH regions to impact product supply, it is highly recommended that the firm contact those regulatory agencies early to alert them to the potential issues and request their assistance.

7.0 Legacy Marketed Products and ICH Q12

One of the challenges with the changing global regulatory landscape relates to marketed products, that is, products that have been approved and are in distribution. The full benefits of the ICH Q12 PAC processes may initially only apply to products that are newly submitted in jurisdictions that apply ICH guidelines. Consequently, for many older products already on the market, there may be a prolonged period to implement ICH Q12 PAC principles, which may be impacted by the inclusion of specific information in the approved filing. For example, identifying which items should be called out as ECs could present a challenge and could require considerable time and resources to accomplish.

To take full advantage of ICH Q12, some type of modernization of the regulatory filing around the CMC sections may be needed.

Regardless of how they embrace ICH Q12 concepts, in ICH regions, firms will have the option to develop and submit a list of ECs to regulatory agencies in a prior-approval submission. This would allow the firm to conduct a full or partial assessment and possibly remove parameters currently in the filing that, based on product knowledge, do not require a submission to change. This could reduce

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the filing category of some of the parameters, for example, moving from prior approval to notification. The firm could also use a prior-approval submission to change from a parameter-based EC to performance-based conditions. This could be done for the entirety of Modules 3.2.S and 3.2.P or only for specific sections.

The potential advantage to adding ECs to a marketed product is that more historical data and a better understanding of the product exists for marketed products than would have been available when the initial marketing authorization was submitted. That product history and knowledge can be used to demonstrate that the changes made to certain parameters and quality attributes are low risk and, therefore, should be allowed notification-only status or could be managed solely within the PQS. To reduce the regulatory burden of PACs, the historic manufacturing and characterization data would have to be submitted to justify the risk assessment scientifically. This data should be submitted in the development sections of the filing (e.g., Modules 3.2.S.2.6 and 3.2.P.2) to support the proposed ECs and reduced filing category. If these data are submitted in sufficient detail, once the ECs are approved, future PACs could be implemented more quickly.

In most cases, prospectively filing a prior-approval submission to add ECs would not be efficient unless it is submitted along with a planned major change. Otherwise, the ECs would have to be approved before a change could be made without needing a submission, which would further increase the regulatory burden. Additionally, regulatory agencies in different jurisdictions may not agree to the same ECs, which would add to the complexity of the maintenance of ECs by the firm.

8.0 Recommendations for Harmonization (Categorization of Changes)

In the current regulatory environment, industry benchmarking shows that it can take up to five years to have a single change approved across all affected countries. Current guidances from regulatory agencies, including the FDA, EMA, WHO, and Health Canada, attempt to clarify how changes are to be evaluated, when they should be reported, and if prior approval from the agency is required to implement the change ***(4, 5, 11, 13, 20)***. Based on the severity of the effect of the change to the ECs or product quality and, hence, to the safety and efficacy for patients, the changes are generally classified in three categories:

- Do and Record (Lowest Risk) — manage within the PQS (e.g., Health Canada Level IV)
- Do and Tell (Minor to Moderate) — notification (e.g., Type IA or Annual Reportable)
  — Tell, Wait and Do (some Moderate) — (e.g., Type IB or CBE-30)
- Tell, Wait and Do (Major) — prior approval (e.g., Type II or Prior Approval Supplement)

The EU's variation regulations and post-authorization guidance and FDA's scale-up and post-approval change (SUPAC) guidance documents provide specific details on the scientific data that should be submitted and how it should be presented for specific PACs ***(27-31)***. The requirements are consistent with the potential impact of the change on the quality, safety, and efficacy of the product similar to the underlying concepts adopted by the FDA in its *Guidance for Industry: Changes to an Approved NDA or ANDA* ***(32).***

Specific scientific documentation requirements and approval timelines can vary greatly among jurisdictions, however.

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To allow the industry to continue to modernize and make the needed improvements in both quality and productivity in a timely fashion, action is needed in many of the non-ICH regions. Establishing new regulations or modifying existing regulations to introduce the three mechanisms for change—PQS management, notification, and prior approval—that align with ICH Q12 and other global regulatory expectations is recommended. To fully realize the benefits of ICH Q12, a certain level of regulatory convergence and harmonization is needed. With respect to categorization of changes, this convergence and harmonization is necessary for global movement towards the model provided in **Figure 3.0-1**.

8.1 Harmonized Definitions and Consistent Requirements for Approval of Changes

The categorization of changes depicted in **Figure 3.0-1** as major, moderate, or minor throughout all regions would greatly benefit both regulators and manufacturers globally by providing clear, consistent guidelines.

Having a consistent set of requirements in place would allow firms to prepare a single scientific justification for their PACs that can be submitted to all regulatory agencies for approval. Specifically, better definition of the requirements for the approval of a major change would clarify data-reporting expectations. With this clarity, firms could better plan for major changes. With this convergence and harmonization using the ICH framework, regulatory agencies also could provide guidelines on the requirements for comparability that include a time-period requirement for stability data for each of the three categories of change. Applying the same requirements across ICH and non-ICH regions would also benefit patients.

8.2 Mutual Recognition and Consistent Review Times of PACs in ICH Regions

Most often, product applications are filed across multiple regions. If the classification of changes and requirements for justification of a PAC were consistent throughout all ICH regions, a firm would be able to submit the approval evidence of a change from one ICH member to other ICH members, minimizing the repetitious review time for subsequent approvals. If member countries mutually recognize the approval of a PAC, filing the change in one ICH member country could then be sufficient to obtain approval in all ICH regions.

Similar concepts could be applied in non-ICH regions. Acceptance of changes approved by recognized regulatory agencies in ICH countries based on sound science- and risk-based principles would result in a lower regulatory burden worldwide. This is an established practice, for example, with issuing Certificates of Pharmaceutical Product. Many jurisdictions accept a Certificate from one jurisdiction and leverage the approval from the Certificate-issuing country to alleviate the regulatory burden in the country accepting the Certificate.

Taking this reasoning further, the review timelines for a PAC submission could be defined by the type of change, which would then be similar throughout the ICH regions and the rest of world. (**Table 8.2-1** provides an example of such a proposal.) Making this change would ensure that firms can better anticipate the timeline for approvals of the PAC, allowing for improved planning, faster introduction of innovative technologies, and timelier availability of improved products to patients.

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Table 8.2-1 Proposed Harmonized Standard Review Time

	Consistent PAC Risk-Level Reporting Category and Review Times		
	Do and Record (Lowest Risk)	Do and Tell – Notification (Minor to Moderate)	Tell, Wait and Do – Prior-approval (Major)
Review Time[§]	Not Applicable, documented within a firm's PQS	Regional timelines apply but NMT 90 days[#]	Timelines are jurisdiction-specific[#*] but NMT 180 days

NMT = No more than

[§] Excludes potential clock stops due to queries

[#] Ranges from implementation upon filing to a specified timeline post-filing; "NMT" timelines are strongly encouraged and recommended here

[*] Timelines for major prior-approval submissions may be impacted by site evaluations or inspections

9.0 Conclusion

PACs and the continuing work to harmonize the processes used by the regulatory authorities to govern them is a key enabler to fully implementing a continual improvement model that ensures access to state-of-the-art, high-quality medicines and prevents or mitigates supply disruptions. The principles outlined in ICH Q12 provide the pathway to harmonization.

If implemented globally, the principles set out in ICH Q12, including PACMPs, can lead to more consistent reporting requirements, with the level of reporting based on the risk associated with a specific change. This, in addition to increased alignment of regulatory authority processes and approval times, can enable a more rapid implementation of innovative technologies and improvements in the manufacturing process.

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11.0 Appendix 1: Replacement of Traditional Adventitious Agent Detection Methods with Rapid, PCR-based Detection Methods

11.1 Background

Changes to adopt new and improved technologies for test methods can pose a challenge to the supply chain as this type of change typically requires regulatory agency filings and, sometimes, prior approval.

The use of testing methods to detect adventitious agents in biologics manufacturing is a necessity and is required by regulators globally. During the manufacturing process for biologics, manufacturers may use eukaryotic expression systems, which can be prone to microbial contamination. Consequently, these processes require strict microbial control strategies, including testing for such adventitious agents as mycoplasma, *Leptospira,* and viruses.

Current adventitious agent test methods, however, pose numerous limitations. Assays based on polymerase chain reactions (PCR) offer the potential to mitigate these limitations and offer even better overall detection of adventitious agents. Biologics manufacturers are actively exploring this area, and current research indicates that PCR-based testing is not only scientifically acceptable, but also accepted by regulators.

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PCR-based test methods for *Mycoplasma sp.* present certain advantages for microbiological control—better sensitivity, faster turnaround of results, reduced sample handling, and fewer test failure issues (e.g., due to sample matrix effects)—and timely batch release for biologic drug substances.

11.1.1 Description and Rationale of Proposed Changes

The proposed change(s) should be described in sufficient detail to enable a regulatory agency to evaluate the relevancy and adequacy of the PAC strategy, such as supporting data and change categorization. Including information on the basis and rationale for the change(s), where applicable, is recommended. The differences from the data that have already been approved should be highlighted clearly (preferably in a tabular format, as illustrated in **Table 11.1.1-1**).

Currently, mycoplasma testing of bulk harvest is performed according to established compendial methods—the culture method (broth and agar cultivation procedure), and the indicator-cell culture method (cultivation using an adequate cell line followed by a DNA-staining procedure) ***(1-2)***.

In this case study, the firm plans to introduce an alternative PCR-based method for the detection of *Mycoplasma sp.* in a product's bulk harvest. This PCR method would be used to perform mycoplasma testing on unprocessed bulk harvest samples as an alternative mycoplasma test method. The existing, registered test for mycoplasma (compliant with both European and U.S. pharmacopeias) will remain in the application.

Table 11.1.1-1 List of Proposed Changes

Current State	Proposed Change	Rationale (see details below)
Traditional mycoplasma detection methods (Ph. Eur. 2.6.7/USP <63> culture and indicator cell culture methods)	PCR-based adventitious agent mycoplasma detection method	• Automated handling of the specimens, avoidance of significant manual handling • Significantly reduced process time—less than 1 day compared to 28 days using the traditional, growth-based and indicator-cell-based mycoplasma detection methods (**Mycoplasma**, **Leptospira**, or viruses could contaminate expensive downstream processing equipment during the traditional 28-day method, resulting in a possible supply disruption) • Fully automated readout—no need for well-trained analysts with a high level of expertise to interpret results • No need for specialized equipment and laboratories to perform microbiological growth-based and indicator-cell cuture-based assays • Less prone to interference—some cytotoxic New Molecular Entities (NMEs) interfere with in vitro cell culture assays • Increased sensitivity—lower dilution needed to overcome product inhibition • Increased specificity • Less complex sample workflow—only PCR-based assays will be used • Fewer safety issues—due to the fully automated sample workflow, avoids manual pipetting that might result in repetitive strain injuries

11.1.2 Present State of Testing

Current detection methods for traditional adventitious agents require the following:

- Significant manual handling of the specimens
- Test period of up to 28 days for results, during which time *Mycoplasma*, *Leptospira*, or viruses could contaminate the downstream processing equipment, resulting in a possible supply disruption

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- Well-trained analysts with a high level of expertise to interpret results
- Specialized equipment and laboratories

Additional drawbacks of traditional cell- and growth-based detection methods for adventitious agents are:

- Some cytotoxic NMEs interfere with in vitro cell culture assays
- Low sensitivity due to dilution needs (see **Table 11.1.2-1**)
- Limited specificity
- Different assay platforms
- Complex sample workflow
- Safety, health, and environmental issues
- Risk of secondary contamination, as some adventitious agents require the use of viable positive controls

Increasing concern has been raised in recent years regarding the sensitivity and/or specificity of traditional detection assays for adventitious agents. Validation studies of the lack of detection limit/sensitivity have demonstrated the superiority of PCR-based rodent parvovirus detection assays over the traditional 324K-cell-based detection assays. **Table 11.1.2-1** presents a summary of the validation results.

Table 11.1.2-1 Summary of Validation Results

Rodent Parvovirus (Species or Strain)	Sensitivity of the 324K-Cell-Based Assay/(TCID50/mL)	Sensitivity of the PCR-Based Assay/(TCID50/mL)
Kilham's rat virus (KRV)	>10	0.003
Mouse Minute Virus (MMV17)	>10	0.14
Mouse Minute Virus (MMV52)	>10	0.27
H1-Parvovirus	0.86	0.02

Regarding the lack of specificity, a biopharmaceutical firm discovered that mouse minute virus (MMV) had contaminated its Chinese hamster ovary (CHO) cell production system ***(3)***. Neither the in vitro general viral screen nor the 324K cell assay detected this MMV strain, though it can be reliably detected using a PCR assay. This clearly shows that not all viruses can be detected by indicator-cell-based in vitro viral screen assays.

In addition, this limited specificity, which translates into potential inaccuracies in detecting contaminants, has been confirmed not only for viral contaminants, but also for bacterial contaminants. For example, *Spiroplasma citrii* cannot be detected by the traditional growth- and cell-based mycoplasma detection assays. **Table 11.1.2-2** provides a summary of validation results ***(4)***.

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Table 11.1.2-2 Detection Limit/Sensitivity of Mycoplasma Detection Assays

Mycoplasma Strain	Reference Strain	Detection Limit Culture Method (CFU/mL)	Detection Limit Indicator-Cell Culture Method (CFU/mL)	Detection Limit Mycoplasma PCR (CFU/mL)
M. hominis	ATCC 23114	0.1	1	0.1
M. orale	ATCC 23714	0.1	Not detectable	1
M. arginini	ATCC 23838	0.1	10	1
A. laidlawii	EDQM	0.01	0.01	0.1
M. hyorhinis	EDQM	1	1	10
M. pneumoniae	NCTC 10119	1	Not detectable	1
M. salivarium	ATCC 23064	0.1	Not detectable	1
M. fermentans	ATCC 19989	1	10	0.1
Sp. citri	ATCC 27556	Not detectable	Not detectable	1
M. hyorhinis[a]	ATCC 29052	Not detectable	10	10

[a] Fastidious Strain

The data summarized in **Table 11.1.2-2** illustrate that the culture method and the mycoplasma PCR assay are sensitive methods for detecting mycoplasma contamination in preharvest cell-culture fluid samples derived from a CHO-cell culture process. Both methods can detect the mycoplasma species included in this comparability study with a sensitivity of ≤10 CFU/mL, but the PCR-based mycoplasma detection method is superior to the traditional mycoplasma detection methods in regard to specificity.

While this limited specificity and detection limit/sensitivity is a current challenge for the development of NMEs, it will become an even greater challenge in the future. Some of the NMEs, such as anticancer therapeutics, interfere with traditional in vitro indicator-cell culture-based detection methods for adventitious agents. The mode-of-action of some NMEs is based on cellular cytotoxicity. Cytotoxicity-induced cell death not only kills the oncogenic target cells but also the permanent cell lines used as indicator cells for the traditional in vitro detection methods. Therefore, the testing sample needs to be diluted (product-dependent up to 1:10,000), resulting in very low sensitivity of the traditional in vitro detection methods. Whether or not this situation will be tolerated by regulators in the future is questionable; however, regulators have published guidance documents recommending PCR-based methods for specific target organisms.

Another example demonstrating the acceptance of PCR-based detection methods by regulators can be found in the FDA's *Q&A on Current Good Manufacturing Practices, Good Guidance Practices***.** Here, the Agency raised the question of whether the microbial control strategies of biologics manufacturers are capable of detecting unusual microbiota such as *Leptospira.* The FDA recommends the "use of validated PCR methods (e.g., as an investigative tool) for rapid screening and detection of spirochete bacteria" ***(5)*****.**

To address the potential limitations of the traditional detection methods discussed in **Section 11.1.2,** several companies have introduced multiyear programs, intending to replace traditional cell- and growth-based detection methods (for microbial contaminants, e.g., mycoplasmal contaminants, or specific virus strains, such as MMV) with new, fully automated PCR-based methods. These new methods include purification of the target nucleic acid and real-time PCR amplification and detection systems.

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Automated PCR-based testing methods for adventitious agents, in contrast to traditional indicator-cell and growth-based methods, detect contaminants in five to eight hours of analytical time. Thus, biopharmaceutical companies can isolate the contamination to upstream and harvest equipment without contaminating downstream equipment. No additional or subsequent processed lots using the contaminated downstream equipment are secondarily contaminated. Consequently, decontamination efforts are less extensive, resulting in reduced facility shut-down times. In general, PCR-based adventitious-agent detection methods significantly reduce reaction times, minimize the impact of any potential contamination on the product supply, and ensure an uninterrupted supply to patients.

Higher throughput and higher reliability are additional benefits of PCR-based methods as they can be executed on fully automated robotic platforms without the need for manual intervention. This avoids safety, health, and environmental hazards, including such ergonomic issues as repetitive strain injury. Other benefits include cost savings, improved productivity, and increased agility and flexibility in the testing network.

11.2 Current Filing Considerations

Given that the regulatory landscape is not globally harmonized, different submission strategies must be applied to introduce PCR-based adventitious-agent detection methods by PACs:

- **U.S.—** Either by product-specific control-system lifecycle-management submissions, multi-product changes-being-effected (CBE-30) submissions, or prior-approval supplements (PAS)
- **EU—** By product-specific analytical variations

The current and future regulatory requirements are summarized in **Table 11.2-1**.

In the current U.S. and EU biologics regulatory guidances, this method change is not defined for a biologic; therefore, it must bear the caveat that the agencies reserve the right to respond with feedback to the proposed comparability protocol and approve, deny, and/or ask for more information.

Table 11.2-1 Summary of Current and Future Regulatory Requirements

		Current Regulatory Requirements			Future Regulatory Requirements	
Existing Established Conditions or Impacted Sections	What's changing?	Supporting Data Required to Demonstrate Equivalence	Additional Tests Required	Current Filing Category	Justification w/ Supporting Data, Testing & Risk Assessment	Proposed Filing Category
Can be an in-process control (IPC) or a release specification • IPC (S.2.2.S.2.4) • Relese specification (S.4.1, S.7.1, S.7.3)	Replacement of conventional mycoplasma testing with more sensitive PCR-based method	See "Commitment and Implementation Plan" in the example below	See details below	• US: PAS (Tell, Wait and Do) • EU: Type IB (Tell and Do)	PACMP as prior approval in each jurisdiction prior to change implementation	Follow up with post-change data as "Notification" of fulfilled commitment (Do & Tell)

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11.2.1 Risk Management

The PCR method has been scientifically proven to be more sensitive and easily able to detect more mycoplasma strains than the 21-day method. The risk of the current, older method having less detectability will be mitigated by replacing it with a more sensitive method. In addition, a risk assessment of the new PCR method was completed, and additional risk controls will be implemented as part of the change plan.

11.3 Comparability Protocol Strategy Overview for the Replacement of the Traditional Mycoplasma Detection Methods with a PCR-based Method

According to Ph. Eur. *Mycoplasma 2.6.7* and Ph. Eur. *Nucleic Acid Amplification Techniques 2.6.21*, the following parameters should be evaluated for a qualitative assay by the validation program ***(1,6)***:

- **Specificity—** ability to unequivocally assess target nucleic acid in the presence of other (potentially interfering) components that may be expected to be present
- **Detection Limit/Sensitivity—** lowest amount of target nucleic acid in a sample that can be detected, but not necessarily quantitated as an exact value
- **Robustness—** measure of its capacity to remain unaffected by small, but deliberate, variations in method parameters and provide an indication of test method reliability during normal usage
- **Comparability Study—** comparison of the detection limits of the alternative method and official methods; however, specificity should also be considered

The validation program could be split into two sections:

- **Generic Method Validation—** evaluation of all the parameters listed above
- **Product-Specific Method Validation—** use of different lots of the specimen to be tested with a limited number of sensitivity experiments to evaluate whether the PCR-based detection method is unaffected by the specific sample matrix (test for interfering or inhibiting properties of the sample matrix or method suitability test) and by lot-to-lot-variability of the sample to be analyzed

The experimental data generated during the generic method validation program, as well as the data of the product-specific method validation(s) method suitability test(s), would be included in the submission package.

11.3.1 Conditions that Need to be Met Prior to Implementation of the Change: Acceptance Criteria for the PCR-Based Mycoplasma Detection Method

11.3.1.1 Acceptance Criteria for the Generic Method Validation

11.3.1.1 Specificity

The ability of the PCR-based mycoplasma detection method to detect a large panel of mycoplasma species should be demonstrated using characterized reference panels (e.g., reference strains). Moreover, as the primers will likely detect other bacterial species, the potential cross-detection should be documented in the validation study.

Based on the results of this validation of specificity, if a gap in the specificity of the method is identified (such as detection of nonmycoplasmal bacteria species), an appropriate strategy must be proposed in the validation study to allow interpretation of positive results on a routine basis. For example, a second test may be performed using an alternative method without this specificity gap or by using an official method.

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11.3.1.2 Detection Limit/Sensitivity

A positive cut-off point should be determined for the PCR-based detection method. The positive cut-off point, as defined in Ph. Eur. 2.6.21, is the minimum number of target sequence copies per volume of sample that can be detected in 95% of test runs.

Acceptance criteria for the detection limit/sensitivity are defined as follows:

- If the alternative method is proposed to replace the culture method, the PCR-based mycoplasma detection system must be shown to detect 10 CFU/mL for each relevant mycoplasma test species chosen.
- If the alternative method is proposed to replace the indicator-cell culture method, the nucleic acid amplification techniques (NAT) system must be shown to detect 100 CFU/mL for each mycoplasma test used.

11.3.1.3 Robustness

To determine whether the assay remains unaffected by deliberate variations caused by different lots of reagents and solutions, three lots of selected critical PCR assay ingredients are used for the PCR-based mycoplasma detection assay.

Using any of these combinations, the PCR-based mycoplasma detection assay must be able to detect at least 10 CFU/mL of any of the mycoplasma reference preparations (e.g., *M. orale*).

11.3.1.4 Comparability Study

In the comparability study, the PCR-based alternative method should be performed in parallel with the official methods to evaluate simultaneously the detection limit/sensitivity of both methods using CHO cultures inoculated with different spike levels of mycoplasma reference strains.

The detection limit/sensitivity, as well as the specificity of the PCR-based mycoplasma detection method, should be comparable or superior to the traditional mycoplasma detection methods.

11.3.1.5 Acceptance Criteria for Product-Specific Method Validation

To evaluate whether the test specimen possesses inhibitory or interfering properties on the PCR-based mycoplasma detection method, the parameter detection limit/sensitivity should be examined with three different lots of the test specimen (method suitability test).

The detection limit/sensitivity of the PCR-based assay determined in the generic method validation must be confirmed using a defined set of mycoplasma species for different lots of the test specimen.

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Example: Post-Approval Change Management Protocol

For Change in Analytical Method

PURPOSE

The intent of this validation is to demonstrate that the mycoplasma PCR assay is suitable for the detection of mycoplasma species in preharvest cell-culture fluid (PHCCF) samples. It also fulfills the requirements given in Ph. Eur. 2.6.7 for an alternative NAT-based method proposed as an alternative to the indicator-cell culture and the culture methods for the detection of mycoplasmas. According to Ph. Eur., the criteria specificity, sensitivity/detection limit, and robustness should be addressed. In addition, a comparability study with the compendial methods, the indicator-cell culture method, and the culture method should be performed.

The presence of mycoplasma DNA in the sample is determined by a 400-500 bp target sequence that is amplified; this amplicon is detected finally by polyacrylamide gel electrophoresis or by real-time PCR (qPCR).

Details of the assays and validation data are published in a peer-reviewed article by Deutschmann, et al.2010 ***(7)***.

This protocol provides guidance on how to validate the PCR-based mycoplasma detection method for:

- Generic validation of the mycoplasma PCR assay. This part of the overall validation is based on the validation requirements of Ph. Eur. 2.6.21 and the ICH Q2(R1) guideline for a qualitative test ***(6,8)***
- Comparability study for method validation between the mycoplasma PCR assay and the two compendial mycoplasma detection methods (culture method and indicator-cell culture method)
- Study on inhibiting substances of the PHCCF to evaluate whether the PHCCF interferes with the performance of the mycoplasma PCR assay

BACKGROUND

Materials and Methods

Test Microorganisms

- *M. hominis* ATCC 23114
- *M. orale* ATCC 23714
- *M. arginini* ATCC 23838
- *Acholeplasma laidlawii* (EDQM reference strain) ***(9)***
- *M. hyorhinis* (EDQM reference strain) ***(10)***
- *M. hyorhinis* ATCC 29052 (fastidious strain, used in comparability study)
- *M. pneumoniae* NCTC 10119
- *M. salivarium* ATCC 23064
- *M. fermentans* ATCC 19989
- *Spiroplasma citri* ATCC 27556

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Nonmycoplasmal Strains

In addition, the following closely related non-Gram-positive mycoplasma strains will be used for specificity experiments:

- *Streptococcus bovis*
- *Lactobacillus acidophilus*
- *Clostridium sporogenes*

CHO Cells

- CHOK1 cell line (DSMZ No.: ACC 110) will be used for the generic method validation and test positive controls (GAPDH PCR)
- PHCCF samples of three different fermentation runs will be evaluated in the study on PCR-inhibiting or PCR-interfering substances

DESCRIPTION OF PLANNED CHANGES

Generic Validation of the Mycoplasma PCR Assay

The validation of the mycoplasma PCR assay follows the criteria described in Ph. Eur. 2.6.21 and in the ICH Q2(R1) guideline for a qualitative test. According to the Ph. Eur., the criteria specificity, sensitivity/detection limit, and robustness should be addressed.

Specificity

The specificity of the assay will be tested with a large panel of the mycoplasma strains listed in the "Materials and Methods" section, as well as with other bacterial species that might be cross-detected in the assay. The bacterial species listed in Materials and Methods were chosen for this study due to the close phylogenetic relationship to mycoplasma strains. CHO cells will be spiked with mycoplasma strains as noted. These spiked cells will be examined as a sample in the mycoplasma PCR assay.

In addition, CHO cells will be spiked with a serial dilution of the Gram-positive bacteria to concentrations of 10^6–10^2 CFU/mL. These spiked cells will also be examined as a sample in the mycoplasma PCR assay.

Acceptance Criteria

Acceptance criteria of the test assay must be met for all positive and negative controls.

An amplicon of 400–500 bp must be detectable for all mycoplasma strains tested.

No acceptance criterion is defined for the cross-detection of the Gram-positive bacteria.

Sensitivity/Detection Limit

The detection limit of the mycoplasma PCR assay is the lowest number of CFU in a sample that can be detected but not necessarily quantitated. To determine the detection limit for each mycoplasma strain, three independent 10fold dilution series will be tested. The mycoplasma strains chosen will represent all strains required in the NAT validation chapter (Ph. Eur. 2.6.21), as well as *M. orale*, *M. salivarium,* and *M. hominis*.

According to the data published by McGaritty et al., the five strains—*M. hyorhinis, M. orale, M. arginini, M. fermentans,* and *A. laidlawii*—account for more than 95% of all isolates found in mammalian cell cultures ***(11)***.

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More recent publications by Timenetsky et al. and Uphoff & Drexler confirm the data published by McGarrity et al. In addition, both *M. salivarium* and *M. hominis* have been described as sources of mycoplasma contamination in mammalian cell cultures. These literature citations support the relevancy of the selected validation strains to industry contamination risks ***(12-14)***.

The determination of the sensitivity/detection limit of the mycoplasma PCR method will be performed using the mycoplasma strain stocks listed in the Comparability Study section below. For the calculation of the different mycoplasma spike concentrations, the titer before freezing will be used.

To determine the detection limit for each mycoplasma strain stock, three independent runs (dilution series) will be prepared on different days with eight test replicates for each dilution, giving a total number of 24 test results. In each dilution series, CHO cells will be spiked with 10, 5, 1, 0.5, and 0.1 CFU/mL. The spiked CHO cells will then be analyzed as samples in the mycoplasma PCR assay. If the cut-off point is reached for one organism, no further testing of the higher dilutions will be evaluated. The cut-off point is the minimum number of CFU/mL of sample that can be detected in 95% of the test runs. If necessary, additional spike concentrations will be used.

Acceptance Criteria

Acceptance criteria of the test assay must be met for all positive and negative controls.

If the mycoplasma PCR assay is proposed as an alternative to the compendial methods (culture method and indicator-cell culture method), the Ph. Eur. claims that the alternative method can detect 10 CFU/mL for each mycoplasma strain mentioned. Therefore, for each of the strains tested for the detection limit, a band of 400–500 bp must be present in no fewer than 23 of 24 test results (=95% cutoff) up to a concentration of at least 10 CFU/mL.

Robustness

In order to examine the robustness of the ready-to-use PCR mycoplasma detection assay, CHO cells will be spiked with *M. orale* master stock at concentrations of 1000, 100, 10, 1, 0.1, and 0.01 CFU/mL. The spiked CHO cells will be analyzed as samples in the PCR mycoplasma detection assay. The tests will be carried out independently from each other by one analyst on three different days.

The PCR mycoplasma detection assay was developed for use with two models of thermocycler. Each thermocycler uses different mechanisms for heating and cooling; hence, both thermocycler models will be included in the robustness study.

To determine whether the assay is affected by different lots, two lots of the commercial kit will be used to perform the experiments.

Acceptance Criteria

Acceptance criteria of the test assay must be met for all positive and negative controls.

A band of 400–500 bp must be present up to a concentration of 10 CFU/mL in all three test runs.

The detection limit determined for all three test runs must be comparable (difference between the test runs of not more than 1 log step).

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Comparability Study

To demonstrate equivalency between the compendial test methods (culture method and indicator-cell culture method) and the mycoplasma PCR assay, comparability studies will be performed using the mycoplasma reference strain stocks.

To demonstrate comparability between the indicator-cell culture method, the culture method, and the mycoplasma PCR assay, 15 mL of a CHO-K1 culture will be spiked with the following mycoplasma strain stocks to a concentration of 100 CFU/mL:

- *M. hominis* ATCC 23114
- *M. orale* ATCC 23714
- *M. arginini* ATCC 23838
- *A. laidlawii* EDQM reference strain ***(15)***
- *M. hyorhinis* EDQM reference strain ***(10)***
- *M. hyorhinis* ATCC 29052 (fastidious strain)
- *M. pneumoniae* NCTC 10119
- *M. salivarium* ATCC 23064
- *M. fermentans* ATCC 19989
- *Sp. citri* ATCC 27556

The spike concentration will be determined using the titer of each of the mycoplasma strain stocks before freezing.

Based on this initially spiked sample (100 CFU/mL), a serial 10fold dilution will be prepared using unspiked CHO-K1 cell culture as diluent, leading to samples spiked with 100, 10, 1, 0.1, 0.01, and 0.001 CFU/mL. The spiking will be performed in triplicate.

After spiking, these samples will be subjected to all three different mycoplasma detection methods in parallel without freezing the samples. From each dilution, 10 mL of this spiked sample will be used in the culture method, 1 mL will be used as the inoculum for the indicator-cell culture method, and 900 µL will be subjected to the mycoplasma PCR.

In this comparability study, the detection limit for each respective method is determined as the lowest spike concentration in which mycoplasma can be detected in all three runs.

The detection limit of the three mycoplasma detection methods should be comparable.

Study on PCR-Inhibiting or PCR-Interfering Substances of the Preharvest Cell-Culture Fluid

To evaluate if the PHCCF of CHO-based manufacturing processes possess inhibitory or interfering properties upon the detection of mycoplasma DNA with the mycoplasma PCR assay, PHCCF samples of three different fermentation runs will be spiked with *M. hyorhinis* and *A. laidlawii* to titers close to the detection limit determined in the generic method validation. *M. hyorhinis* has been chosen for this study as one representative of strains with results in a lower sensitivity (5 CFU/mL) in the generic method validation. *A. laidlawii* is chosen because this strain possesses the most mismatches to the primer sequences as compared to all the strains included in the generic method validation.

Sensitivity/Detection Limit

The detection limit for *M. hyorhinis* and *A. laidlawii* will be analyzed in the presence of three different batches of PHCCF samples.

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Therefore, three independent dilution series will be prepared with these two test microorganisms on different days with eight test replicates for each dilution, giving a total number of 24 test results. In each dilution series, one of three PHCCF samples will be spiked with 10, 5, and 3 CFU/mL of *A. laidlawii* or 0.1, 0.05, and 0.01 CFU/mL of *M. hyorhinis*. The spiked PHCCF will be analyzed using the mycoplasma PCR assay.

Acceptance Criteria

Acceptance criteria of the test assay must be met for all positive and negative controls.

To confirm the results of the generic method validation, a band of 400–500 bp must be present in 23 of 24 test results up to a concentration of, at least:

- 10 CFU/mL for the detection of *M. hyorhinis*
- 0.1 CFU/mL for the detection of *A. laidlawii*

COMMITMENT AND IMPLEMENTATION PLAN
Proposed Reporting Category for Proposed PACs

The firm's PQS will be followed to implement the change, and the relevant regulatory authorities will be notified. Post-change monitoring will occur to ensure no unintended consequences result; all documentation relevant to the change will be available for review by regulatory authorities during inspection.

Implementation of Changes

The firm will outline the global implementation plan for this change (e.g., timing of the introduction).

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12.0 Appendix 2: Introduction of a New Isolator System for Aseptic Fill

12.1 Background

Appendix 2 describes the introduction of improved technology for aseptic filling and the related assessment of the impact to currently approved biological products. This example refers to the upgrade from a legacy conventional-filling operation to an isolator technology for toxic, biologically active, genetically modified, cell-based, and/or protein therapeutics. The introduction of the new isolator aims to enhance the quality of the product and reduce microbial, particulate, and chemical contamination as well as increase the sterility safety level and/or operator-safety environment in the case of active biologics or genetically modified organisms.

12.2 Current Regulatory Filing Considerations—Improvements to Drug Product Aseptic Filling

12.2.1 U.S. Filing Considerations

With respect to certain biologics, the FDA *Guidance for Industry: Chemistry, Manufacturing, and Controls Changes to an Approved Application: Certain Biological Products* suggests that changes impacting the transfer of manufacturing of an aseptically processed sterile drug substance or drug product into a new or renovated aseptic processing area within the currently licensed building would be considered a prior approval supplement (PAS). Likewise, the addition or modification to an approved filling line that affects sterility assurance would be considered a PAS ***(1)***.

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In the 1997 FDA *Guidance for Industry: Changes to an Approved Application: Biological Products,* "Section II. Changes under §601.12(b) – Changes requiring supplement submission and approval prior to distribution of the product made using the change (major changes)," item #11 discusses potentially related changes:

> Major construction, or changes in location, involving or affecting environmentally controlled manufacturing or related support areas, such as ***
> - aseptic processing areas;
> - modifications to support systems with significant potential to affect air, water, or steam quality;
> - installation of a new HVAC system involving or affecting environmentally controlled manufacturing or related support areas; or
> - modifications to an existing HVAC system that supplies aseptic processing areas ***(2)***.

In general, the process impact of changing the drug product filling process to replace the filling unit and introduce an isolator system would be considered a major change and would require a PAS (four-month review). Both the footprint of the existing filling suite and the HVAC system will be impacted. The needed space may be reduced, as an isolator would have a Grade A chamber that can be placed in a Grade C or D space, making a Grade B background no longer necessary. In addition, other adaptations would be needed to accommodate material and airflow accessibility.

FDA offers general information about aseptic processing in *Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing – Current Good Manufacturing Practice* and sets high-level expectations for industry in *Guideline for the Submission of Documentation for Sterilization Process Validation in Applications for Human and Veterinary Drug Products*. Both documents summarize the types of data the FDA and other regulatory authorities look for in aseptic filling processes; however, neither provides specific guidelines about isolators or types of equipment ***(3, 4)***.

12.2.2 Canadian Filing Considerations

The Health Canada *Guidance Document: Post-Notice of Compliance (NOC) Changes: Quality Document,* "Appendix III, Biologics, 3.2.P.3 Manufacture," places the following changes in the "Notifiable Change" reporting category:

> 40. *Change involving a drug product manufacturer/manufacturing facility*, such as: *** b. replacement or addition of an equivalent formulation/filling suite, and ***
>
> 42. *Change in the drug product manufacturing process*, such as: *** b. addition or replacement of equipment [e.g., formulation tank, filter housing, filling line and head, and lyophilizer] within the existing filling areas****

A notifiable change involves a seven-day screening and logging period, plus a 90-day target review ***(5)***.

The same guidance places the following changes in the supplemental new drug submission (SNDS) reporting category:

> Appendix III, Biologics, 3.2.P.3 Manufacture
>
> 40. *Change involving a drug product manufacturer/manufacturing facility*, such as: ***c. addition of a non-equivalent formulation/filling suite to an approved formulation/filling facility****
>
> Appendix I, Drugs, 3.2.P.3 Manufacture
>
> 25. *Change in the drug product manufacturing process****

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An SNDS (CMC only) involves a 45-day screening and logging period, plus a 180-day target review.

A separate filing is required for each product impacted; however, data from one filing may be leveraged for other products if, for example, a range of vials and fill volume is considered.

12.2.3 EU Filing Considerations

According to Commission Regulation (EC) No 1234/2008 *Guideline on the details of the various categories of variations to the terms of marketing authorizations for medicinal products for human use and veterinary medicinal products* [B.II.b.3, b], variations related to "substantial changes to the manufacturing process, … which may have a significant impact on the quality, safety or efficacy of the medicinal product" are classified as a major variation of Type II. That includes variations to the manufacturing process, formulation, specifications, or impurity profile of an active substance or finished medicinal product. Such major variations require approval of the relevant regulatory agency before implementation ***(6)***.

12.2.4 WHO Considerations

According to WHO *Annex 4 – Guidelines on Procedures and Data Requirements for Changes to Approved Vaccines* [TRS No. 993], changes in the manufacturing process of a final product, such as the addition or replacement of equipment, including filling lines, are indicated as a "moderate quality change".

Implementation of changes in the moderate (or major) category requires reporting to the national regulatory agency to supplement the information in the original marketing authorization or product license. Major and moderate quality changes should be reviewed and approved by the regulatory authority prior to implementation of that change ***(7)***.

This aligns with WHO *Annex 3 – Guidelines on Procedures and Data Requirements for Changes to Approved Biotherapeutic Products* [TRS No. 1011], which proposes that a change in a filling line would be a moderate quality change that should be reviewed and approved by the national regulatory authority prior to implementation ***(8)***.

12.3 Strategy

In this example, the current regulatory requirements versus the proposed requirements, leveraging ICH Q12 principles ***(9)***, are summarized in **Table 12.3-1**.

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Table 12.3-1 Change in Filling Technology

		Current Regulatory Requirements			Future Regulatory Requirements	
Existing Established Conditions or Impacted Sections	**What's changing?**	**What supporting data are required to demonstrate equivalence?**	**Any additional tests (e.g., analytical, stability, validation) required other than existing?**	**Current Filing Category**	**Justification, including Supporting Data and Testing & Risk Assessment**	**Proposed Filing Category**
Manufacturing Process Development (3.2.P.2.3) Description of Manufacturing Process and Process Controls (3.2.P.3.3) Controls and Critical Steps and Intermediates (3.2.P.3.4) Process Validation and/or Evaluation (3.2.P.3.5) Batch Analyses (3.2.P.5.4) Stability (all sections within 3.2.P.8) Facilities and Equipment (3.2.A.1)	Replacement of Conventional Filler with New Isolator	See "Commitment and Implementation Plan" in the example below	See detail below	US: PAS (Tell, Wait and Do) EU: Type II (Tell, Wait and Do) CAN: Notifiable Change or SNDS (Tell, Wait, and Do)	Provide PACMP as prior approval in each jurisdiction prior to change implementation (Tell, Wait and Do)	Follow up with post-change commercial lot data as Notification of fulfilled commitment (Do and Tell)

The strategy summarized in **Table 12.3-1** is intended to leverage the use of a post-approval change management protocol (PACMP, or comparability protocol) consistent with ICH Q12 in order to modify the filing category for changes based on prior agreement between a firm and a regulatory agency. The PACMP will aim to demonstrate to regulatory authorities how the proposed improvement in the filling manufacturing process generally affects the relevant quality attributes of the product in order to demonstrate that the planned modifications do not adversely impact the final product quality, safety, and efficacy, as stated in ICH Q5E ***(10)***.

Acceptance of the PACMP would be subject to a prior approval review (four-month review, depending on jurisdiction), but may allow for shorter review periods for each product subsequently affected (i.e., every product that is filled in an impacted suite).

In addition, requesting a presubmission meeting to obtain initial buy-in with the agency should be given consideration.

In this example of upgrading from conventional filler to isolator, consideration for processing capabilities between the previously used equipment compared to the proposed equipment may be very informative.

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Post-Approval Change Management Protocol

For Introduction of a New Isolator

PURPOSE

This post-approval change management protocol (PACMP) describes the strategy and approach for demonstrating the introduction of an isolator as an improvement to an aseptic filling suite for Ourfavoritebiologic (OFB).

This protocol outlines the proposed studies, test methods, and acceptance criteria that will be used to assess the impact on OFB drug product when manufactured using the new isolator. The data from the proposed studies will demonstrate that the quality of the OFB drug product is at least comparable to the historical OFB drug product and that the new isolator is acceptable for manufacturing this product.

SCOPE OF CHANGE

Sterile drug product manufacturing is accomplished in various ways; traditionally, ISO-5 laminar air flow (LAF) units with manual operations have been widely used. As the sterility of the product is of the greatest concern, manual operations always have disadvantages over isolated systems. The overall objective in the aseptic filling process is to attain the highest sterility assurance level by using an isolated system. Currently, this site uses ISO-5 LAF conventional filling technology, an open restricted access barrier system (RABS). To switch from LAF to any other isolated system for the filling of protein therapeutics as sterile solutions may require an extensive overhaul of the manufacturing process; however, the increased control over the following critical areas (in order of importance) plays a major role in maintaining the quality of the product ***(11)***:

- Microbial contamination
- Particulate contamination
- Chemical contamination

Isolated systems available in the industry to exercise that control are:

- Closed RABS (access by glove ports)
- Isolators (access by glove ports with automated decontamination)
- Aseptic work cell with robot (gloveless systems)

Critical quality attributes (CQAs) and critical process parameters (CPPs) affected by any fill–finish process are illustrated in **Figure 12-1** using a fishbone structure.

A thorough assessment was completed with the three options (i.e., RABS, isolator, aseptic work cell) weighted against product impact, production needs, facility capabilities, and business needs. Moving forward with the isolator technology was deemed best suited for meeting the needs of the business and ensuring product quality.

The filling unit in the suite (i.e., conventional filler using LAF) used for production of OFB will be completely replaced with new equipment, including the introduction of an isolator. As part of this change, the filling line would clearly separate the influences of the environment from the processing. The new equipment will have a new pump (current filling uses a diaphragm pump; the proposed filling will use a peristaltic pump that may be capable of increased output and greater filling accuracy than the cur-

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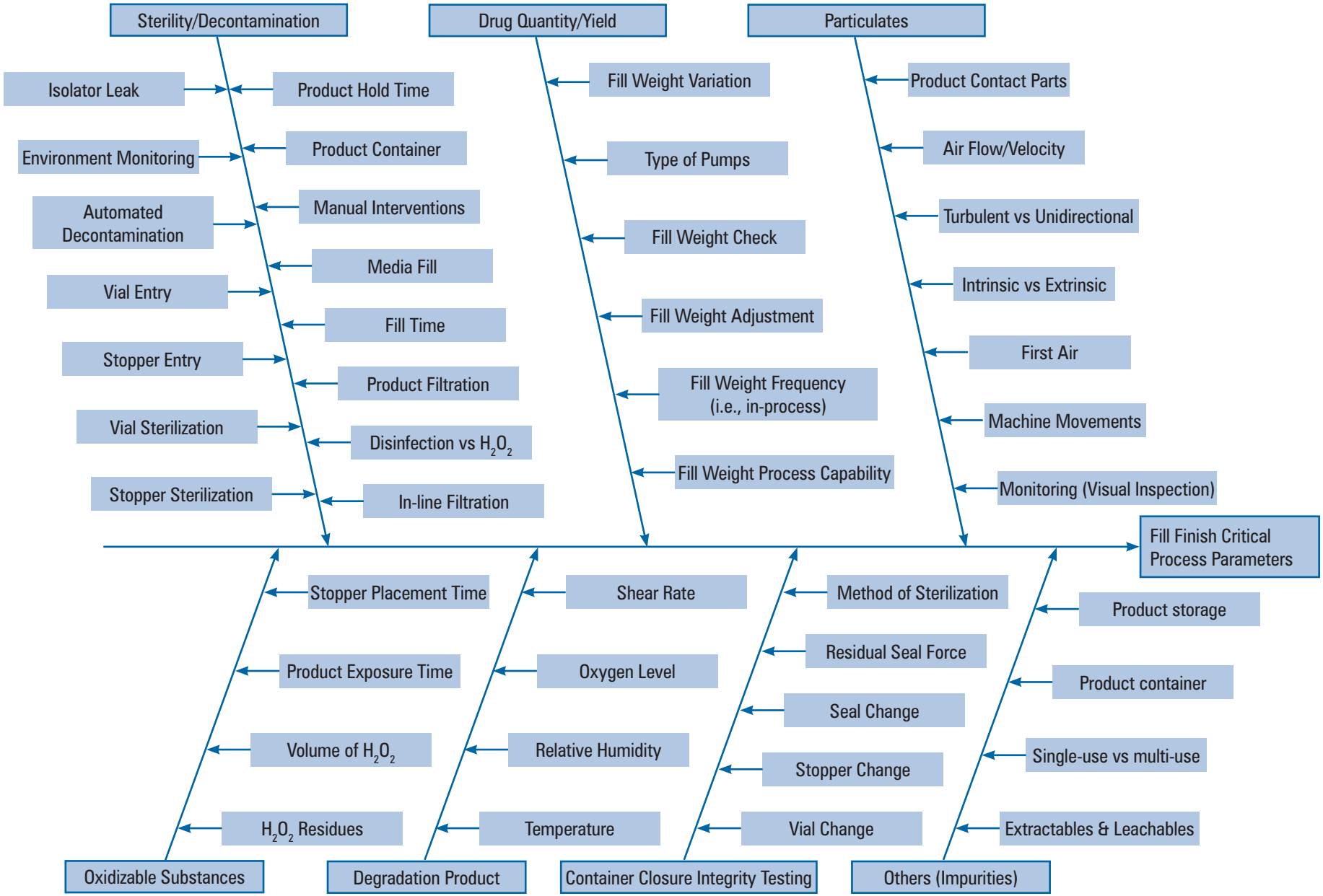
Figure 12-1 Considerations and Critical Quality



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rently approved filling line. The new filling isolator will require modifications to the facility, including the HVAC system and personnel and material flow, and will improve product isolation systems. This is beneficial as automated integration will, in turn, provide additional process control and consistency, minimizing human intervention, and provide the highest sterility assurance for the filling operations.

No change to change-over procedures, visual inspection procedures, release specification, or cleaning procedures is proposed because of this new isolator. All of these are described in this PACMP. The proposed isolator introduces automated decontamination via vapor phase hydrogen peroxide (H_2O_2) cycle between campaigns to decontaminate the isolator chamber. Both viable and nonviable particle monitoring will be conducted inside the isolator. Further details will be reported in the process performance qualification (PPQ).

To support the changes introduced by a new filling line, the following information is provided in this proposed PACMP:

- Background and currently approved process and procedures
- Risk and product impact assessments
- Equipment used for the new isolator will be comparable to that used in the current aseptic filling suite; specifically, a stainless-steel shell and frame for the isolator will continue to be used and all tubing will still be silicone. The new pump, however, could potentially impact the product quality due to shear forces. A laboratory study was performed to evaluate the pump and establish its operating parameters. These studies demonstrated the new pump will not negatively impact the product quality and identified the filling parameters. The pump settings identified in these studies were used in the Engineering runs. Information on the Laboratory and Engineering studies is provided later in this PACMP.
- Summary of CQAs and CPPs and plans for PPQ
- Justification for the change

All changes to the process, equipment, and facility will be qualified as described in the PACMP prior to implementing manufacture of OFB.

BACKGROUND

Current Drug Product Manufacturing

The aseptic manufacturing process for OFB consists of sterile filtration, aseptic filling, capping or closing, inspection, labeling, packaging, and storage of the finished drug product. The filling areas used for OFB are shared with other sterile liquid products. All products are manufactured on a campaign basis so that only one product is present in the manufacturing or filling suite at a time. A validated changeover procedure is implemented prior to the initiation of a new campaign.

All OFB product-contact manufacturing equipment is dedicated to OFB. Product-contact materials are primarily stainless steel and silicone tubing, and other product-contact materials are glass and acrylics. Elastomers used in the manufacturing process (e.g., gaskets, tubing) meet the specifications of USP Class VI, as defined by USP <88> Biological Reactivity, In Vivo, and 21 CFR 177.2600 – Rubber Articles Intended for Repeated Use ***(12, 13)***.

Lot Size(s)

There are two lot sizes for OFB: 450 L and 1250 L. The yield can vary from 50,000–225,000 vials and 135,000–625,000 vials, respectively. All products filled on this filling line use the same lot sizes and are supported by media fills spanning these ranges.

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Fill Process

The currently approved process is provided in **Table 12-1**.

Table 12-1 Current Process

Additions/Removals	Steps	Manufacturing Parameters	In-Process Tests
Filling Suite X			
Bulk drug substance Samples removed for QC testing	Sterile filtration	Pressure: ≤20 psi Product Volume to Surface Area (V:SA): ≤175 mL/cm^2 Post-use filter-integrity bubble point: ≥50 psi	Pre-filtration bioburden: ≤10 CFU/mL
▼			
Filling Suite – Aseptic Core (Grade A) – No Isolator			
	Filling	Line speed: ≤60 vials/min Weight check-in intervals of ~5% of target batch size	Environmental monitoring
▼			
	Stoppering	Fully stoppered vials	Environmental monitoring
▼			
Capping Room			
Samples removed for QC testing	Capping	Capped vials with integral seals	Environmental monitoring Finished product testing (Release testing)
▼			
Packaging Suite			
	Packaging		

Currently approved CQAs and CPPS are provided in **Table 12-2** and **Table 12-3,** respectively. The proposed process is provided in **Table 12-5**.

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Table 12-2 Current Critical Quality Attributes

Critical Step	Analytical Method	Acceptance Criteria
Pre-sterile Filtration	Bioburden	≤10 CFU/mL
Product Release Criteria		
Identification	Conforms to reference standard	Meets criteria
Appearance	Visual	Solution is clear to colorless to yellow-green; there may be a few particulates.
Potency	Biological Assay	75-125%
Purities/Impurities	HPLC and GC	Meets criteria
Particulates (subvisible)	Light Obscuration (LO)	≥2 μm – N/A ≥5 μm – N/A ≥10 μm – ≤6000 particles/container ≥25 μm – ≤600 particles/container
pH	pH meter	7.6 ± 0.2
Bacterial Endotoxin	Turbidimetric LAL	<5 EU/mL
Sterility	Sterility	Meets 21 CFR 610.12
Excipients: Polysorbate 80	Assay	0.100 ± 0.050 mg/mL

Table 12-3 Current Critical Process Parameters

Process Step	Manufacturing Criteria	Proven Acceptable Range
Sterile Filtration	Pressure test: Post-use filter-integrity bubble point following WFI flush	≥50 psi
Filling (Filling uniformity and homogeneity)	Vial weight checks: Weight check in intervals of ~5% of target batch size	±1.5%

Primary Packaging

OFB is packaged using a standard pharmaceutical container–closure system, specifically, Type 1 clear glass vials and bromobutyl rubber stoppers, known to be safe and compatible with liquid protein products. The same container–closure components are used for several sterile protein products on this filling line as they do not interact and cause changes in the quality of the product.

Table 12-4 provides a general description of the materials used in the primary container–closure system. No changes are proposed to the currently approved primary container–closure system due to the proposed changes to the filling line.

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Table 12-4 Primary Container Closure System Used on Filling Line

Material	Description	Dimensions/Sizes
Type I glass (not sulfate or sulfate-treated)	Meets: • USP <660> • Ph. Eur. 3.2.1	3 ml, 6 ml, and 10 ml
Bromobutyl rubber stopper (pre-siliconized)	Meets: • USP <381> • Ph. Eur. 3.2.9 (Type I closures) • ISO 8871	20 mm
Aluminum seal	—	—
Polypropylene and resin cap	—	—

Washing/Sterilization of Components

No changes are proposed to the currently approved washing and sterilization of components due to the proposed changes to the filling line.

Vials

The processing of vials used for filling OFB and all drug products filled in this line consists of:

- Visual Inspection—100% of vials are visually inspected
- Sampling—Defined number of vials from any lot are sampled according to vial specifications to be released for use in filling
- Cleaning—Vials for manufacturing use are washed by a validated cycle on the vial washer
- Sterilization—Washed vials are heat-sterilized and depyrogenated using a depyrogenation oven, then fed through a dry-heat tunnel
- Filling and Closure—Sterilized vials are filled, stoppered, and capped

Stoppers

Stoppers are provided prewashed and siliconized. Stoppers are steam-sterilized and stored in sealed containers until required for use.

Container Closure Integrity Testing

When new container–closure components or a change to the filling process that may impact container closure are introduced, container–closure integrity assessments are performed. In order to confirm comparability following the introduction of the isolator, container–closure integrity will be tested to support this change. Container and closure system integrity testing includes performing validated physical or chemical container and closure system integrity tests (e.g., bubble tests, pressure/vacuum decay tests, trace gas permeation/leak tests, dye penetration tests, seal force or electrical conductivity tests, and capacitance tests) or microbiological container and closure system integrity tests (e.g., microbial challenge or immersion tests). Such tests may be more useful than sterility testing in demonstrating the potential for product contamination over the product's shelf life or dating period. The container and closure system integrity tests for OFB will be designed to detect a breach of the container and/or closure system prior to product contamination.

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Currently Approved Gowning Procedures

Personnel entering the manufacturing areas gown and degown according to specified requirements following approved written procedures. Different levels of gowning occur according to the environmental controls and to the type of operation being carried out in the manufacturing rooms. Gowning garments might include safety shoes, shoe covers, hair cover, facial hair cover (as needed), safety glasses, coveralls, booties, sterile gloves, sterile sleeves, or sterile mask in line with written procedures. Personnel are required to wear gloves prior to handling any cleaned, sanitized, or in-process production equipment.

An external laundry service may be used for cleaning and/or sterilization of reusable coveralls and uniforms, where applicable.

Cleaning Procedures

The facility is cleaned and sanitized using primary disinfectants and sporicidal agents at defined intervals to control mold, bacteria, and spore-forming microorganisms. All manufacturing areas are cleaned and sanitized as required in written procedures, and all cleaning and sanitization activities are documented. The firm periodically reviews the data obtained from environmental monitoring to determine if further modifications to the facility cleaning program are warranted. Bioburden is verified by two testing methods: Contact surface samples and airborne samples for viable particulates.

Change-Out Procedures

The currently approved and the proposed filling line will be used on a campaign basis across several protein therapeutics, all filled into glass vials in a multiproduct facility. Filling needles, gloves, all tubing, connectors, and glass containers are single-use and replaced for each campaign. The manufacturing area is dedicated to the production of only one product at any one time, with complete cleaning and changeover between products.

Throughout the manufacturing process, appropriate measures are established to ensure the absence of product contamination. Other methods used to prevent contamination include facility design features, environmental monitoring, dedicated equipment, room air classification, air pressure differential, flow of personnel, materials and waste, and cleaning procedures.

With respect to the currently approved and proposed filling process, there are no reusable direct product-contact materials. Multiuse product-contact equipment is cleaned following validated processes, and the cleaning effectiveness is verified by total organic carbon methods. Process equipment is sterilized into the aseptic core or proposed isolator by autoclave sterilization or dry-heat depyrogenation processes.

The product-contact components in the formulation (carboys, stir bars, sampling materials, filters, and transfer tubing) are sterilized prior to use and are single-use, disposable materials. All product-contact components in the filtration system (tubing and filters) and in the filtrate receiving-vessel (glass carboy, tubing, and carboy ports) are sterilized prior to use and are single-use, disposable materials.

The currently approved process uses a diaphragm pump, and proposed filling lines use a peristaltic pump system. The product-contact components (tubing, connectors, and filling nozzles) are sterilized prior to use and are single-use, disposable materials. Parts of the filling machine and other equipment that cannot be sterilized by thermal methods are sanitized by wiping external surfaces with approved and validated germicidal or sporicidal agents. Such chemical sanitization procedures are restricted to equipment that cannot be sterilized by other means and to equipment that is not in direct product contact or in the direct process path.

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Media Fill Program

A media fill is one part of the validation of an aseptic manufacturing process. Media fills are performed every six months and after any changes to the aseptic fill suite are made. Using a sterile microbiological growth medium in place of the drug product, media fills are designed to simulate the filling process to ensure the performance of the aseptic manufacturing procedure. Microbiological growth medium is used in place of the drug solution during media fills to test whether the aseptic procedures are adequate to prevent contamination during actual drug production.

The media fill program is designed to evaluate the aseptic assembly and operation of the critical (sterile) equipment, qualify the operators and assess their technique, and demonstrate that the environmental controls are adequate to meet the basic requirements necessary to produce a sterile drug by aseptic processing. Data collected from media fills are compared to predefined acceptance criteria and run against an approved protocol. The media fills cover the vial sizes specified for OFB (i.e., 3 mL, 6 mL, and 10 mL), ensuring that both the smallest and largest sizes are subject to the media fill.

To assess the impact of the new isolator on the filling suite, media fills will be completed with the new equipment according to the procedures described here and as part of the comparability and PPQ assessment of this change. The viable monitoring requirements during the filling period and the microbial monitoring results for the media fills pre- and post-change will be provided.

DESCRIPTION OF PLANNED CHANGES

A new isolator will be introduced into the manufacturing suite used for the manufacture of OFB. When the new isolator is introduced, the filling line will also be completely replaced using a new pump that has greater throughput and accuracy.

Isolator

The isolator is a "Separative Device" that will provide assured protection by using physical and dynamic barriers to create separation between operation and operator; it comes supplied with HEPA-filtered air and will be decontaminated in a quantifiable and highly reproducible manner. When the system is closed, it uses only decontaminated interfaces ***(11, 14)***.

This change will introduce new technology: a stainless steel–glass isolator and positive pressurization. This design facilitates both aseptic and containment applications; however, it will be used primarily in aseptic processing with the goal of excluding contaminants from the isolator. Construction materials will be mechanically and chemically resistant (i.e., stainless-steel frame, glass). The operator interface will require gloves, gauntlets, and appropriate gowning. The aseptic core will continue to be a HEPA-filtered air supply with vertical, unidirectional air flow (open) and pressure differentials of no more than 100 Pa. Pressure differentials between the exterior and the interior of the isolator will be monitored continuously and integrated into the existing automated process control system.

Sterilization-in-place will be implemented for contact parts, such as fluid pathways. In areas where this cannot be performed, all parts will be autoclaved to achieve sterilization and aseptically assembled prior to processing. Parts of the filling machine and other equipment that cannot be sterilized by thermal methods are sanitized by wiping external surfaces with approved and validated germicidal or sporicidal agents. Such chemical sanitization procedures are restricted to equipment that cannot be sterilized by other means and to equipment that is not in direct product contact or in the direct process path.

A full description of the proposed isolator will be available in the PPQ protocol.

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The scope of this change is limited to the isolator, which consists of a sterilization tunnel, filling machine, and capping machine that will have the capability of processing 3 mL, 6 mL, 10 mL vials of sterile, liquid, injectable drug products. Autoclaves and support utilities are excluded from the scope of this change. (See **Table 12-6** for the impact to process.) The PPQ will be designed according to USP <1208> and will be validated before its use in sterility testing as part of a batch release ***(15)***. To verify that the isolator system and all impacted equipment are suitable for aseptic processing, the following validation studies will be completed: Factory acceptance testing (FAT), site acceptance testing (SAT), installation qualification, operational qualification, and performance qualification. All changes will be assessed in the PPQ, which will confirm that all identified CPPs and CQAs of the drug product are met.

The current manufacturing process is described in **Table 12-1**. The proposed process is provided in **Table 12-5**. Further description of the manufacturing filling process CQAs (**Table 12-2)** and CPPs (**Table 12-3)** are provided in the **Fill Process** and **Validation** sections.

Table 12-5 Proposed Process

Additions/Removals	Steps	Manufacturing Parameters	In-Process Tests
Filling Suite X			
Bulk drug substance Samples removed for QC testing	Sterile filtration	Pressure: ≤20 psi Product Volume to Surface Area (V:SA): ≤175mL/cm2 Post-use filter- integrity bubble point: ≥50 psi	Pre-filtration bioburden: ≤10 CFU/mL
▼			
Filling Suite – New Isolator			
	Filling	Line speed: ≤60 vials/min Target Fill Volume: Target Fill is product- and strength-dependent Weight check-in intervals of ~1% of target batch size	Environmental monitoring
▼			
	Stoppering	Fully stoppered vials	Environmental monitoring
▼			
Capping Room			
Samples removed for QC testing	Capping	Capped vials with integral seals	Environmental monitoring Finished product testing (Release testing)
▼			
Packaging Suite			
	Packaging		

Table 12-6 highlights information about the manufacturing and filling of the product OFB and the proposed changes. The changes will be assessed as per predefined PPQ, taking into consideration the CQAs, CPPs, and lifecycle knowledge management of the product and process.

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Table 12-6 Process Change Impact Assessment

Description (in order of operation)	Change Impact
Manufacturing Site	Same filling suite will be used; however, gowning areas and space will be reduced. Proposed isolator chamber is Grade A, proposed into a Grade C space. HVAC changes, air flow), material flow, waste flow, and personnel flow changes will be captured in PPQ.
Processing Rooms and Equipment (product-contact and non-product-contact) are sanitized	No change proposed
Environmental Monitoring sampling for equipment set-up is initiated	No change proposed
Sterile Equipment is set up; sterile filtration is started	Will be confirmed via validation as part of change
Environmental Monitoring is initiated for filling operations	Will be confirmed via validation as part of change
Vials are transferred from sterilization tunnel into vial feed	Sterilization tunnel is a new feature; washed and depyrogenated vials will be fed through the sterilization tunnel
Target Fill	May be less based on new-found efficiencies; will be confirmed via validation as part of change in equipment
Nominal Fill Volume	No change proposed; will be confirmed via validation as part of change
Stoppers are loaded	No change proposed
Caps are loaded	No change proposed
Sterile, empty Vials will be fed through filling machine with 100% weight check Filled Vials will be stoppered and capped or sealed automatically	No change proposed
100% Visual Inspection of filled vials per USP <790>	No change proposed
Hold Times	Will be confirmed via validation as part of change
Processing Times	No change proposed; however, will be confirmed via validation as part of change in equipment
Batch Size(s)	No change proposed; more vials may be available based on reduced target fill and new-found efficiencies. Will be confirmed via validation as part of change in equipment

Consistent with ICH Q12, the sections of the Common Technical Document (CTD) that will be assessed to evaluate the impact to ECs are listed in **Table 12-7** ***(9)***.

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Table 12-7 Impacted CTD Sections and/or Established Conditions

CTD Section	Section Title	Established Conditions
3.2.P.2.3	Manufacturing Process Development	Update based on final PPQ
3.2.P.3.3	Description of Manufacturing Process and Process Controls	Update based on final PPQ
3.2.P.3.4	Controls and Critical Steps and Intermediates	Update based on final PPQ
3.2.P.3.5	Process Validation and/or Evaluation	For more details on PPQ parameters being assessed, see "Process Validation" CPPs to be assessed will include: Sterilization tunnel: Conveyor speed and time, temperature, pressure, airflow Filler: Conveyor speed, pump operation, tubing Capping: Conveyor speed, compression Formulation: Filtered bulk drug product homogeneity, filtered bulk drug product hold time Sterile filtration: Filter validation Filling: Drug product filling uniformity (fill volume) CQAs to be assessed will include: Sterilization tunnel: Bioburden and endotoxin Filler: Bioburden and weight Capping: Container closure integrity Product: Final drug product release specifications
3.2.P.5.4	Batch Analysis	Release data, as per currently approved drug product specifications for three historical commercial lots compared to three commercial lots post-change
3.2.P.8.1	Stability Summary and Conclusions	At least six months of stability data from three Engineering runs under nominal, accelerated, and stressed conditions Propose to put at least the first lot from PPQ run into the long-term stability program. Real-time data available at time of filing will be provided. Product has a three-year shelf life at 2–8 °C.

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(Table 12-7 continued)

CTD Section	Section Title	Established Conditions
3.2.P.8.2	Stability Commitment	Commitment to continue stability studies for first three PPQ runs and Engineering runs per currently approved storage conditions and shelf life (0, 3, 6, 9, 12, 24, and 36 months at 2–8 °C, and accelerated conditions for 3 months). Thereafter, at least one batch per year will be placed into an ongoing long-term stability program.
3.2.P.8.3	Stability Data	See Section 3.2.P.8.1.
3.2.A.1	Facilities and Equipment	Updates to the facility and suites involved in the manufacture of the drug product, specifically, current state vs proposed new state with new filling equipment and impact on the following, will be provided: Complete set of floor plans and flow charts (drawings, room classifications, water systems, HVAC systems), as appropriate and/or as impacted Cleaning validation, as appropriate Results of the environmental monitoring studies in classified areas Cleaning procedures (including data in a summary validation report) demonstrating lack of carry-over or cross-contamination for new filling process if impacted by new equipment Information describing the change-over procedures for the shared product-contact equipment, if impacted by new filling process in line with new equipment

RISK ASSESSMENT AND MITIGATION PLAN

Risk Assessment

The risk of introducing the new filling system was evaluated; two aspects were considered in the risk assessments:

- A facility design and fit assessment were completed taking into consideration the changes to the facility to allow for installation and commissioning of the new equipment and process for sterile manufacturing. This risk assessment included an overall cGMP assessment, including an evaluation of the personnel, material, equipment, waste flows, classifications, differential pressurization, gowning and associated gowning facilities, materials, and finishes.
- Product and process assessment was completed to assess the product and process CQAs and CPPs impacted by aseptic fill-finish processing as illustrated in **Figure 12-1**. The parameters

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that were assessed as potentially having a significant impact on product quality were identified, triaged for resourcing, and further studied to achieve a higher level of process understanding. An extensive analysis, using a Failure Mode and Effects Analysis (FMEA) tool, was completed as per ICH Q9. The FMEA provided an evaluation of potential failure modes for the aseptic filling process and their likely effect on outcomes and/or product performance. Identifying these failure modes allowed plans to be developed for risk reduction to eliminate, contain, reduce, or control the identified potential failures. Additionally, the FMEA aided in prioritizing the most important risks and their mitigations.

The output/results of the risk assessments were used as a basis for design of the isolator, clean rooms, and utilities and to identify where further analysis and mitigation may be required as described in the Mitigation Plan.

Mitigation Plan

Any potential risk identified with the introduction of these changes will be carefully assessed through the process validation and product comparability assessments described in this PACMP.

To inform the design of the comparability protocol, the potential impact of the change on CPPs (**Table 12-14**) for the filling process and/or CQAs (**Table 12-15**) of the drug product was further evaluated. This evaluation included the data generated and gained from historic product experience and data collected from Engineering runs and Laboratory pump studies summarized in **Table 12-9**. This is not an all-inclusive list of risks identified, only those that will be mitigated by the comparability protocol and/or process validation associated with this PACMP to support this change. The full risk assessment is in the firm's PQS. A risk level was calculated using a risk priority number (RPN) by multiplying the scoring of each risk factor (RPN = Severity × Occurrence × Detection), as seen in **Table 12-8**:

Table 12-8 Risk Level Calculation Table

Risk Level	RPN Range
Low	1-20
Medium	24-45
High	48-125

Table 12-9 Evaluation of Risk to Product and/or Project to Inform Comparability Study Design

Parameter	Risk Level (H, M, L)*	Rationale	Mitigation to Support Comparability
Product Attributes			
Potency	L	Although filling equipment is being improved, the filling process is not changing. Further, primary container closure components remain unchanged. It is very unlikely that the potency would be impacted by this change based on preliminary data collected through Engineering runs.	Potency will be confirmed to meet specification as a part of normal batch release and as per comparability protocol.

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(Table 12-9 continued)

Parameter	Risk Level (H, M, L)*	Rationale	Mitigation to Support Comparability
Product Attributes			
Identity	L	Although filling equipment is being improved, the filling process is not changing. Further, primary container closure components remain unchanged. It is very unlikely that the identity would be impacted by this change based on preliminary data collected through Engineering runs.	Identity will be confirmed to meet specification as a part of normal batch release and as per comparability protocol.
Purity/ Integrity, Product and Process Impurities, Sterility	M	Although filling equipment is being improved, the filling process is not changing. Primary container closure components are not changing. It is very unlikely that the purity profile would be impacted by this change based on preliminary data collected through Engineering runs.	Purity will be confirmed to meet specification as a part of normal batch release and as per comparability protocol. PPQ will assess filling process capability against predefined CPPs, including bioburden and filter integrity. Impact of residual H_2O_2 will also be assessed. Further, the sterile filtration step will be validated, and the prefiltration bioburden and post-use filter-integrity bubble point are predefined CPPs.
Other Attributes			
Yield	M	Although filling equipment is being improved, the filling process is not changing. Primary container closure components are not changing. In addition, the filling process will be validated. Data from Engineering runs support that impact to yield will not be adverse.	PPQ will assess filling process capability against predefined CPPs and will look at yields.
Fill Volume	M	Although filling equipment is being improved, the filling process is not changing. Primary container closure components are not changing. In addition, the filling process will be validated. Data from Engineering runs support that impact to fill volume will not be adverse.	PPQ will assess filling process capability against predefined CPPs and will look at yields.
Container Closure Integrity	M	Although filling equipment is being improved, the filling process is not changing. Primary container closure components are not changing. In addition, the filling process will be validated. Data from Engineering runs support that impact to yield will not be adverse.	Container–closure integrity testing will be completed. Specifically, microbial ingress challenge will be useful to demonstrate the potential for product contamination over the product's shelf life or dating period. The container and closure system integrity tests for OFB will be designed to detect a breach of the container and/or closure system prior to product contamination post-filling change.

* H = High, M = Medium, L = Low

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PROCESS DEVELOPMENT AND CHARACTERIZATION STUDIES
Laboratory Pump Studies

The equipment used for the new isolator will be comparable to that used in the current aseptic filling suite. The new pump could potentially impact the product quality due to shear forces. This laboratory study was performed to evaluate the pump and establish its operating parameters; the result is summarized here. These studies demonstrated that the new pump will not negatively impact the product quality and identified the filling parameters. The pump settings identified in these studies were used in the Engineering runs.

The new isolator will use peristaltic filling pumps (currently a diaphragm pump system is used). A laboratory-scale study, using a pump representative of the one used for the new isolator, was performed to optimize the filling set-up and pump parameters and to evaluate the performance of the pumps about fill volume accuracy, precision, and impact on product quality.

Studies were initially conducted using the OFB placebo solution to identify the pump settings that would optimize the fill volume accuracy and precision. These conditions were then tested using OFB drug product solution to ensure that the addition of OFB does not affect those parameters and to evaluate its quality. The equipment and optimized set-up of the pump configuration resulted in enhanced process control capability for the fill volume compared to the pumps and pump configuration used in the current aseptic fill suite. Further optimizations of the pump settings were carried out during Engineering runs.

To assess the potential impact of shear forces on product quality during the filling, the same OFB drug product solution was pumped through the pump set-up, including filling needles, three times. After the third pump cycle, samples of the OFB drug product solution were analyzed against the CQAs. In addition, testing was performed to evaluate the generation of ≥2 µm and ≥5 µm subvisible particles. This data generated from the proposed peristaltic pump was compared to data generated from the currently approved diaphragm pump.

The data for the pump studies, which demonstrate that the pump settings will not impact OFB product quality, are shown in **Table 12-10**.

Table 12-10 Testing Results for OFB Drug Product after Three Pump Cycles

Test	Acceptance Criteria (Release and Stability)	Results
Appearance	The solution is clear to colorless to yellow-green. There may be a few particulates.	Pass
Identification	Conforms to reference standard	Pass
Potency	75–125%	102%
Purity and Impurities	Meets	HPLC GC
Bacterial Endotoxin	<10 EU/mL	<5 EU/mL
Sterility	Sterile	Sterile
pH	7.6 ± 0.2	7.6

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(Table 12-10 continued)

Test	Acceptance Criteria (Release and Stability)	Results
Subvisible Particle Testing		
≥2 μm	For information only	3587 particles/container*
≥5 μm	For information only	975 particles/container*
≥10 μm	≤6000 particles/container	134 particles/container
≥25 μm	≤600 particles/container	12 particles/container
Polysorbate 80	0.100 ± 0.050 mg/mL	0.090 mg/mL

* Assessed against data generated on product produced with diaphragm pump and shown to have a highly similar subvisible (i.e., ≥2 μm and ≥5 μm) profile

Engineering Runs

Seven Engineering runs were conducted with the new filling equipment, as per a predefined protocol. All vial sizes (3 mL, 6 mL, 10 mL) were run. **Table 12-11** shows the conditions used for each run.

Table 12-11 Filling Conditions for Engineering Runs

Run	Batch Size	Fill Target	Solution
Engineering Run 1	750 L	5 mL (Mid)	Placebo*
Engineering Run 2	1250 L	5 mL (Mid)	Placebo*
Engineering Run 3	750 L	5 mL (Mid)	OFB
Engineering Run 4	750 L	2 mL (Low)	Placebo*
Engineering Run 5	1250 L	2 mL (Low)	OFB
Engineering Run 6	1250 L	9 mL (High)	Placebo*
Engineering Run 7	750 L	9 mL (High)	OFB

* Same physical characteristics of OFB (static and dynamic viscosity)

Engineering Runs 1 and 2 were performed using a placebo solution. The filling conditions identified in the laboratory studies were used for Engineering Run 1. The average fill volume was 6.03 mL with a standard deviation of 0.08 mL. Slight adjustments were made to the pump parameters prior to Engineering Run 2 to improve the precision.

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Figures 12-2 and **12-3** show the fill volume and distribution, respectively, of each vial in Engineering Run 2. The average fill volume in Engineering Run 2 was 6.01 mL, which is comparable to the accuracy observed in Engineering Run 1. The precision was improved in Engineering Run 2, and the standard deviation of the fill volumes was reduced to 0.03 mL. These parameters were used for Engineering Run 3, which was performed with OFB drug product solution.

Figure 12-2 Individual Vial Fill Volumes for Engineering Run 2

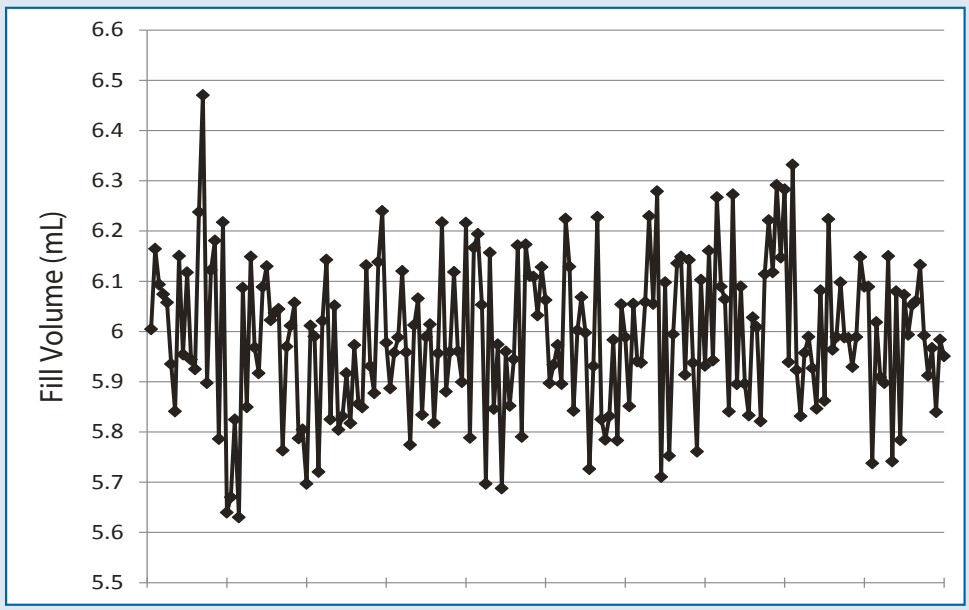
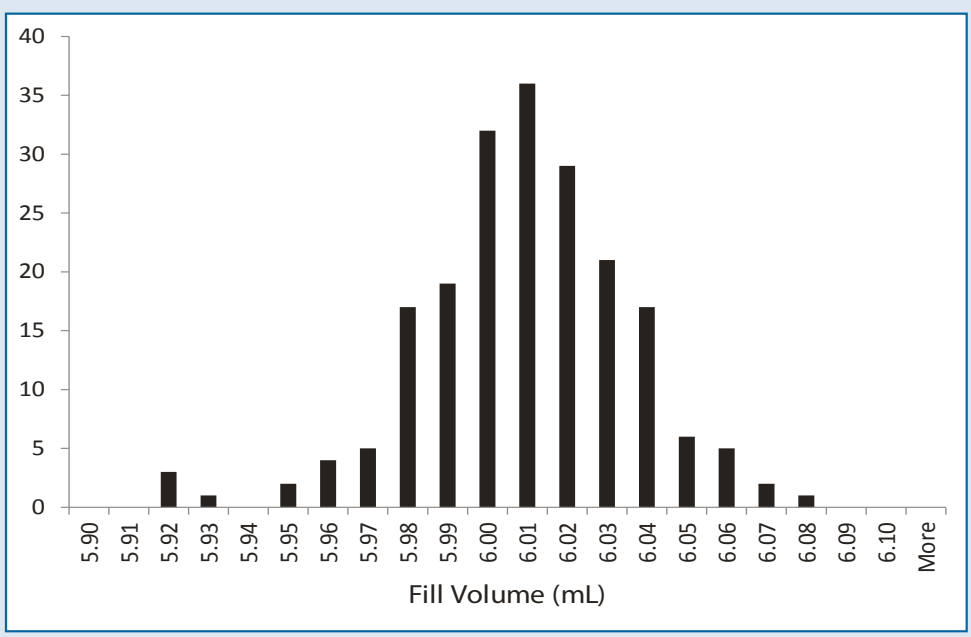


Figure 12-3 Distribution of Fill Volumes for Engineering Run 2



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Engineering Runs 4 and 5 evaluated the lower fill-volume range, whereas Engineering Runs 6 and 7 evaluated the higher fill-volume range. These runs used the same pump parameters that were identified in Engineering Run 2. The data for all Engineering Runs is presented in **Table 12-12**.

Table 12-12 Filling Conditions for Engineering Runs

Run	Fill Volume Target	Average Fill Volume ± 1 Standard Deviation
Engineering Run 1	5 mL	5.00 ± 0.11 mL
Engineering Run 2	5 mL	5.00 ± 0.05 mL
Engineering Run 3	5 mL	4.99 ± 0.04 mL
Engineering Run 4	2 mL	2.02 ± 0.2 mL
Engineering Run 5	2 mL	2.00 ± 0.01 mL
Engineering Run 6	9 mL	9.03 ± 0.10 mL
Engineering Run 7	9 mL	8.99 ± 0.11 mL

Samples from the three Engineering runs that used OFB drug product solution were analyzed against the CQAs, and additional testing was conducted to evaluate the generation of ≥2 µm and ≥5 µm subvisible particles. The data for these studies are shown in **Table 12-13**. These data demonstrated that the filling process does not impact the OFB product quality.

Table 12-13 Testing Results for OFB Drug Product Engineering Runs

Test	Acceptance Criteria (Release and Stability)	Eng Run 3 (6mL)	Eng Run 5 (3mL)	Eng Run 7 (10mL)
Appearance	The solution is clear to colorless to yellow-green; there may be a few particulates.	Pass	Pass	Pass
Identification	Conforms to Reference Standard	Pass	Pass	Pass
Potency	75-125%	102%	99%	115%
Purity and Impurities	Meets	Meets	Meets	Meets
Bacterial Endotoxin	<10 EU/mL	<5 EU	<5 EU	<5 EU
Sterility	Sterile	Sterile	Sterile	Sterile
pH	7.6 ± 0.2	7.6	7.5	7.6
Subvisible Particle Testing				
≥2 µm	N/A	1958	917	3742
≥5 µm	N/A	443	234	719
≥10 µm	≤6000 particles/container	58	26	94
≥25 µm	≤600 particles/container	2	0	4
Polysorbate 80	0.100 ± 0.020 mg/mL	0.112	0.107	0.098

The data from the development studies and Engineering runs demonstrated that the filling equipment used for the new isolator can successfully fill vials of three different sizes with acceptable accuracy and precision. The physicochemical analysis of the OFB product demonstrated that the new filling equipment does not negatively impact the quality of the product.

Comparability Plan and Process Validation

Based on the assessments, the change to the filling process was assessed by both a comparability study and a full-process validation (including pre- and post-change media-fill data completed for the decom-

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missioning of the suite for equipment change-out and required construction and subsequent recommissioning) and the appropriate equipment qualifications.

Process Validation

A clear validation/qualification plan will encompass the following:

- Integration plan for process automation system
- Validation and qualification of the sterilization tunnel
- Equipment and room qualifications (installation and operation qualification plans, equipment longevity assessment/plans, assessment of product-contact equipment and material compatibility/suitability)
- Air flow visualization (smoke test) pre- and post-upgrade
- Media fill pre- and post-change
- Process validation (taking note of how the filling process is impacted by new equipment and space), hold times, residual H_2O_2 assessment, and product impact assessment
- Assessment of process change on facility with respect to environmental monitoring, HVAC systems, material flow, personnel flow, and air flow
- Assessment of impact on established and validated cleaning procedures
- Assessment of impact of new equipment on product container–closure components and assessment of compatibility
- Assessment of impact to existing batch records, work procedures, among other factors
- Container–closure integrity

Process validation for each fill size will be completed under predefined validation PPQ protocols.

Critical Process Parameters

As defined by ICH Q8(R2), a CPP is a process parameter whose variability has an impact on a CQA and, therefore, should be monitored or controlled to ensure the process produces the desired quality ***(16)***.

The CPPs for the filling of OFB are provided in **Table 12-3**. For this PPQ, the CQAs and consideration for lifecycle knowledge, with respect to the proposed isolator, will be assessed as outlined in **Table 12-14**.

Table 12-14 Critical Process Parameters for OFB Drug Product Filling

Process Step	Manufacturing Criteria	Proven Acceptable Range	Justification for Criteria
Sterilization Tunnel	Conveyor speed/time	To be determined	As per Factory Acceptance Testing and manufacturer's validation guide
	Maximum stop time after filling process interruption	To be determined	
	Conveyor temperature and heat distribution in tunnel and glass variation of vials	To be determined	As per Factory Acceptance Testing and manufacturer's validation guide
	Conveyor pressure and airflow	To be determined	As per Factory Acceptance Testing and manufacturer's validation guide

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(Table 12-14 continued)

Process Step	Manufacturing Criteria	Proven Acceptable Range	Justification for Criteria
Filler	Conveyor speed	To be determined	As per Factory Acceptance Testing and manufacturer's validation guide
	Pump operation	To be determined	As per Factory Acceptance Testing and manufacturer's validation guide, peristaltic pumps can be sensitive to tubing changes; will be assessed over time and location
Sterile Filtration	Pressure test: Post-use filter-integrity bubble point following WFI flush	≥50 psi	As per filter manufacturer's validation guide
Filling (Filling uniformity and homogeneity)	Vial weight checks: Weight check-in intervals of ~1% of target batch size, sampling from each of the filling pumps/nozzles in sequence	±1.5%	The filling accuracy was validated for the dedicated 3 mL, 6 mL, and 10 mL (respectively) pump assembly by performing three trials at the minimum volume of 20 mL, and three trials at the maximum volume of 2 mL, 5 mL, and 9 mL, using 3 mL, 6 mL, and 10 mL vials respectively. The check weight system operates by removing one vial prior to fill by robotic fingers; the tare weight is recorded and the vial is then returned to the fill line, filled, and removed by robotic fingers at the gross weight station. The filling accuracy is determined based on the programmed target fill weight.
Hold Times	Drug product maximum hold times will be confirmed via PPQ	To be determined	Controls for this step include: The tank is stored at 2-8 °C prior to the bulk drug being filled. Bacterial and endotoxin samples are taken prior to final sterile filtration and filling.

Critical Quality Attributes

As defined by ICH Q8(R2), a CQA is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality.

Importantly, there are no changes proposed that impact the primary, secondary, or tertiary packaging. A description of the primary container closure system is provided in the "Container Closure" section. As stated, this is a sterile liquid product; no compounding takes place.

Current and proposed inputs are as follows (see also "Container Closure"):

- Glass vials: 3 mL, 6 mL, and 10 mL
- Stoppers: Bromobutyl rubber stoppers
- Caps: Aluminum seal, flip-off
- Product filled at ambient temperature, but tank held at 2-8 °C prior to filling; hold times are validated.

Current and proposed outputs are as follows (see also "Container Closure"):

- Washed vials from vial washer

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- Depyrogenated vials
- Filled vials within weight tolerance and 100% weight check from the filling machine
- Fully stoppered vials
- Capped vials with integral seals

The associated CQAs for OFB are summarized in **Table 12-2**. For this PPQ, the CQAs and consideration for lifecycle knowledge, with respect to the proposed isolator, will be assessed as outlined in **Table 12-15**.

Table 12-15 Critical Quality Attributes and Lifecycle Knowledge Management

Critical Step	Analytical Method	Acceptance Criteria
Pre-sterile filtration	Bioburden	≤10 CFU/mL
Sterilization Tunnel and Filler	Bioburden and endotoxin	Meets criteria
Container Closure Integrity	Microbial ingress challenge	Meets criteria
Product Release Criteria		
Identification	Conforms to reference standard	Meets criteria
Appearance	Visual	The solution is clear to colorless to yellow-green; there may be a few particulates.
Potency	Biological Assay	75-125%
Purities Impurities	HPLC and GC	Meets critera
Particulates (subvisible)	Light Obscuration (LO)	≥2 μm – N/A ≥5 μm – N/A ≥10 μm – ≤6000 particles/container ≥25 μm – ≤600 particles/container
pH	pH meter	7.6 ± 0.2
Bacterial Endotoxin	Turbidimetric LAL	<5 EU/mL
Sterility	Sterility	Meets 21 CFR 610.12
Excipients: Polysorbate 80	Assay	Meets criteria

The results from the first three commercial lots of OFB will be assessed against data from the currently approved commercial process (three historical lots). Comparability will be assessed against the preset specifications described in **Table 12-15**. Comparability acceptance criteria have been determined based on product and process knowledge and data. For quantitative release assays, where applicable, a statistical approach for determination of comparability specifications will be used to demonstrate that the introduction of the new isolator has no impact on identity, strength, quality, purity, or potency of the product.

Container–Closure Integrity Testing

Microbial challenge testing will be executed according to the guidance outlined in *PDA Technical Report No 27: Pharmaceutical Package Integrity*. The PDA recommends the following critical test parameters:

- Challenge media formulation: Tryptic soy broth (TSB) media will be used, which is acceptable for the cultivation of aerobic microorganisms
- Challenge organism: *Pseudomonas aeruginosa* as referenced in PDA TR-27 as a suitable challenge organism

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- High viability count in challenge media at end of test (≥10^5 CFU/mL): 10^6 CFU/mL will be used
- Differential pressure exposure to simulate product processing conditions: No extremes of pressure or vacuum are applied during product processing, therefore, no pressure or vacuum will be applied
- Ability of sterile-packaged media to support growth of challenge organism: Growth promotion will be performed to verify that the media can support microbial growth
- Sensitivity of analytical method if used to detect growth: Visual inspection will be performed to detect growth
- An exposure time of 1–2 hours and incubation period of 14 days are proposed

Challenge organisms recommended are *Escherichia coli*, *Serratia marcesans*, *Clostridium sporogenes*, *P. aeruginosa*, *Staphylococcus epidermidis*, and *Brevundimonas diminuta* ***(17).***

All three vial sizes (3 mL, 6 mL, and 10 mL) will be tested.

Cleaning Validation

The PPQ will capture all cleaning validations through the new filling process.

Isolator Chamber Decontamination

The isolator system ensures a defined reduction of microbial contamination on the surfaces of the isolator chamber using hydrogen peroxide. H_2O_2 decontamination cycles for the isolator in the syringe and vial lines were developed by defining the parameters that ensured a sterility assurance level of 6-log microbial reduction.

For the decontamination cycle, *Geobacillus stearothermophilus* with a population of at least 10^6 pores/carrier are placed at defined locations, representing worse-case locations, within the isolator chambers.

An assessment of potential impact to product and exposure to low levels of ozone and/or hydrogen peroxide will be completed.

Media Fills

Three media fills (aseptic process simulation) will be performed to requalify the filling process, the equipment, and the aseptic process following the introduction of the isolator. The media fills will be executed under protocol and serve to validate the aseptic process as a whole, using the sterile components, controlled room, and filling equipment to fill sterile media. The media fill validation will be simulated at all steps: from introduction of components from the sterilization tunnel to sterile filtration of the drug substance (using media), to preparation of the containers and closures and the filling, stoppering, and sealing of the drug product (media). The three media fills will bracket all vial sizes (3 mL, 6 mL, and 10 mL), container sizes (neck diameters), fill speeds, and filling time durations.

Air Visualization Studies

Smoke studies, or airflow visualization tests, will also be conducted during the qualification of the new isolator for aseptic filling. The smoke studies are conducted to confirm airflow direction in the cleanroom.

Full details of all validation and qualification exercises will be included in the PPQ protocol.

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The purpose of this study is to determine the impact of the introduction of improved isolator technology to the filling suite for drug OFB (and all drugs filled on the same line). The drug product filling changes impact Site PDACorp.; the suite footprint remains unchanged. Drug product (at the commercial scale) filled using the new technology will be compared to currently approved drug product to confirm comparability as per ICH Q5E ***(10)***. The data collected and analyzed under the Engineering runs and Laboratory pump studies described above have demonstrated that the introduction of the new isolator will yield highly similar drug product, and no adverse impact on the safety or efficacy profile of the final drug product is foreseen. This will be confirmed through this comparability protocol and integration and evaluation of all data collected, including routine batch analyses (leveraging the currently approved release parameters to complete physiochemical comparability), in-process control, process validation/evaluation data, container–closure integrity testing, and stability. The comparison of the results to the predefined criteria will allow for an objective assessment of whether the pre- and post-change products are comparable.

Comparability testing comprises release testing, which is summarized in **Table 12-16**.

Table 12-16 Release Testing

Standard Claimed (e.g., Professed, House, USP, Ph. Eur.)		
Specification, Reference Number, and/or Version		
Test	Acceptance Criteria (Release and Stability)	Analytical Procedure (Type/Source/Version)
Appearance	The solution is clear to colorless to yellow-green. There may be a few particulates.	Visual Inspection
Identification	Conforms to Reference Standard	IR Spectrum
Potency	75-125%	Biological Assay
Purity and Impurities	E.g., identified impurities, unidentified impurities and total impurities	HPLC GC
Bacterial Endotoxin	≤5 EU/mL	Turbidimetric LAL
Sterility	Sterile	Compendial
pH	7.6 ± 0.2	pH meter
Excipients: Polysorbate 80	0.100 ± 0.050 mg/mL	Assay
Subvisible Particles ≥10 μm ≥25 μm	 ≤6000 particles/container ≤600 particles/container	Light Obscuration (LO)

Additional characterization data (including completion of container–closure integrity testing) is not informative to this isolator change as the primary container closure has not changed. Further, the process parameters of the new filling equipment will be validated, and the CPPs and CQAs will be confirmed accordingly. A summary of the filling process parameters impacted by this change is provided in **Table 12-17**.

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Table 12-17 Summary of Changes from Currently Approved Process to Proposed Process

Parameter	Current Process	Proposed Process
Product Formulation	L-histidine Polysorbate 80 Maltose	Same
Container Closure	Vial: Type I (not sulfate or sulfate-treated) glass (siliconized)	Vial: Same
	Stopper: Bromobutyl rubber stopper (pre-siliconized)	Stopper: Same
	Flip cap: Aluminum Seal with Flip-off Cap	Flip cap: Same
	Glass vials are washed and depyrogenated, batch-style, in an oven validated for this purpose	Glass vials are washed and depyrogenated in a validated tunnel
In-Process Controls	Prefiltration bioburden	Prefiltration bioburden
	Environmental monitoring	Environmental monitoring
Filling	Conventional filler	Isolator
	Fill Volume Limits Target Fills: 3 mL vials: 2 mL 6 mL vials: 5 mL 10 mL vials: 9 mL Alert Limits: 3 mL vials: 1.9–2.1 6 mL vials: 4.7–5.3 mL 10 mL vials: 8.5–9.5 mL	Same
Capping	Reject vials with raised stoppers, 100% visual inspection	Same
Process Hold Times	A maximum hold time from drug substance to filling is seven (7) days at 28 °C in a stainless steel vessel	Hold times will be validated during process validation

Testing and Acceptance Criteria

The testing and acceptance criteria to be used to assess comparability pre- and post-introduction of the isolator are as follows:

- Three filled commercial-scale lots executed against PPQ (1 lot per vial size, PPQ will include visualization study results pre- and post-change), media-fill results (pre- and post-change), and confirmation of fill volume via extractable fill-volume study
- Filter validation to be completed
- Hold times to be successfully validated
- Cleaning procedures to be successfully validated
- Same three lots also to meet the release specifications outlined in **Table 12-16**.
- Confirmed results from container closure integrity test execution with a commitment to follow through to the end of shelf life

Justification of Change

The introduction of the state-of-the art isolator for aseptic filling is considered necessary to enhance the current GMP standing. Currently, the site utilizes a conventional filler with an aseptic Grade A core within a suite with a Grade B background.

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Contamination may take the form of particulates, microorganisms, or endotoxins. Human operators, as a source of both viable and nonviable particles, pose the greatest risk to product contamination during aseptic processing. Introducing the isolator filling technology significantly reduces these risks:

- Number of process operation steps, which will reduce the potential for an error that ultimately could lead to the distribution of a contaminated product
- Need for intervention(s), that is, any manual or mechanical manipulation of the sterilized drug, components, containers, or closures prior to or during aseptic assembly that poses the greatest risk of contamination

The isolator will also create a physical barrier, lending protection to the product, container, and closure, preventing exposure to the environment (i.e., currently "open" process steps in the LAF filler). Use of the improved filling system will further enhance protection of the product from human contact.

Upon approval of this PACMP and with successful execution of the studies (see "Comparability and Process Validation"), introduction of the proposed filling line is considered a process improvement. All data will be provided post-change as a Notification.

Commitment and Implementation Plan

PDACorp will submit the final study report for this PACMP and the proposed PPQ upon successful completion of the studies described.

The following sections or established conditions will be updated and submitted with the results of the PACMP and PPQ in the follow-up Notification submission:

3.2.P.2.3 Manufacturing Process Development – Including comparability final report.

3.**2.P.3.3 Description of Manufacturing Process and Process Controls** – Description of the manufacturing process, highlighting differences from the approved process, and information on the controls performed at critical steps of the manufacturing process of the proposed drug product.

3.2.P.3.5 Process Validation and/or Evaluation – Process validation and/or evaluation studies (e.g., equipment qualification, media fills) as described, following the PACMP and planned PPQ protocol. All information demonstrating the qualification of the equipment (operational qualification, performance qualification) or qualification of the change will be provided as final reports.

3.2.P.5.4 Batch Analysis – Batch analysis of lots filled with previously approved equipment compared to lots filled after the change.

3.2.P.8.1 Stability Summary and Conclusions – Based on data available at the time of filing.

3.2.P.8.2 Stability Commitment – Commitment to continue stability studies for the first three PPQ and Engineering runs in accord with currently approved storage conditions and shelf life (0, 3, 6, 9, 12, 24, and 36 months) at 2-8 °C; in addition, thereafter, to place at least one batch per year on stability.

3.2.P.8.3 Stability Data – At least six months of stability data from three Engineering runs under nominal, accelerated, and stressed conditions. Real-time data available at time of filing will be provided for the first PPQ lot.

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3.2.A.1 Facilities and Equipment – Information on the proposed production facility involved in the manufacture of the drug product, specifically, current state versus proposed new state with new filling equipment, and impact on the following will be provided: Complete set of floor plans and flow charts (drawings, room classification, water systems, HVAC systems), as appropriate and/or as impacted, as well as the cleaning validation, as appropriate. In addition, the results of the environmental monitoring studies in classified areas will be provided. Cleaning procedures (including data in a summary validation report) demonstrating lack of carryover or cross-contamination for new filling process if impacted by new equipment, information describing the change-over procedures for the shared product-contact equipment, if impacted by the new filling process as per new equipment, as applicable.

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